The Cerebrum and Cerebral Cortex

Department Anatomy

Neuroanatomy (2)

Second Part

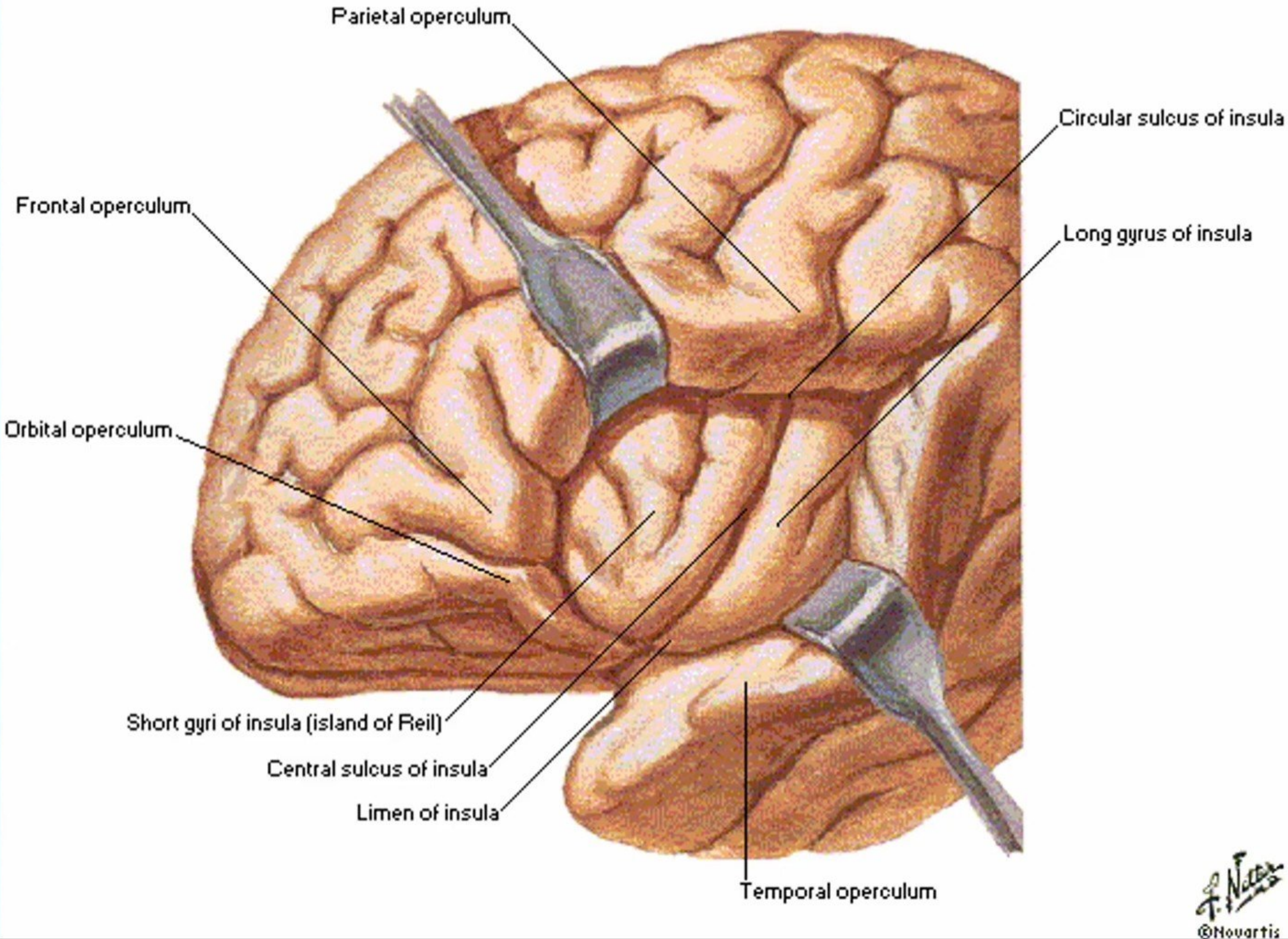
Dr. osman Ahmed Osman

Lectuer Points:

- **Embryonic (developmental) divisions of the Brain.**
- **Division of the forebrain vesicle into the telencephalon and the diencephalon.**
- **The basic components of the CNS.**
- **Cerebrum – Cerebral Hemisphere**
- **Structure of the cerebral cortex.**
- **Layers of the Cerebral Cortex.**
- **Nerve Fibers of the Cerebral Cortex.**
- **Cerebral Features. (The cerebral hemispheres)**
- **Lobes of the Cerebral Cortex. (Sulci , Gyri , Functional Areas & Lesions)**
  - Frontal lobe.
  - Parietal lobe.
  - Temporal Lobe.
  - Occipital lobe.
- **Insula.**
- ***Limbic Lobe (The limbic system).***
- **Basal Nuclei.**
- **White Matter of the Cerebral Hemispheres.**
- **Lobes of Cerebral Cortex and Associated Integration.**
- **Motor homunculus on the precentral gyrus.**
- **Functional Areas of the Cerebral Cortex**
- **Domenet hemospher.**
- **Association areas.**
- **Probable nerve pathways involved in reading a sentence and repeating it out.**
- **Probable nerve pathways involved with hearing a question and answering it.**

Insula

(often called **insula**, **insulary cortex** or **insular lobe**) is a portion of the cerebral cortex folded deep within the **lateral sulcus** (the fissure separating the temporal lobe from the **parietal** and **frontal lobes**).



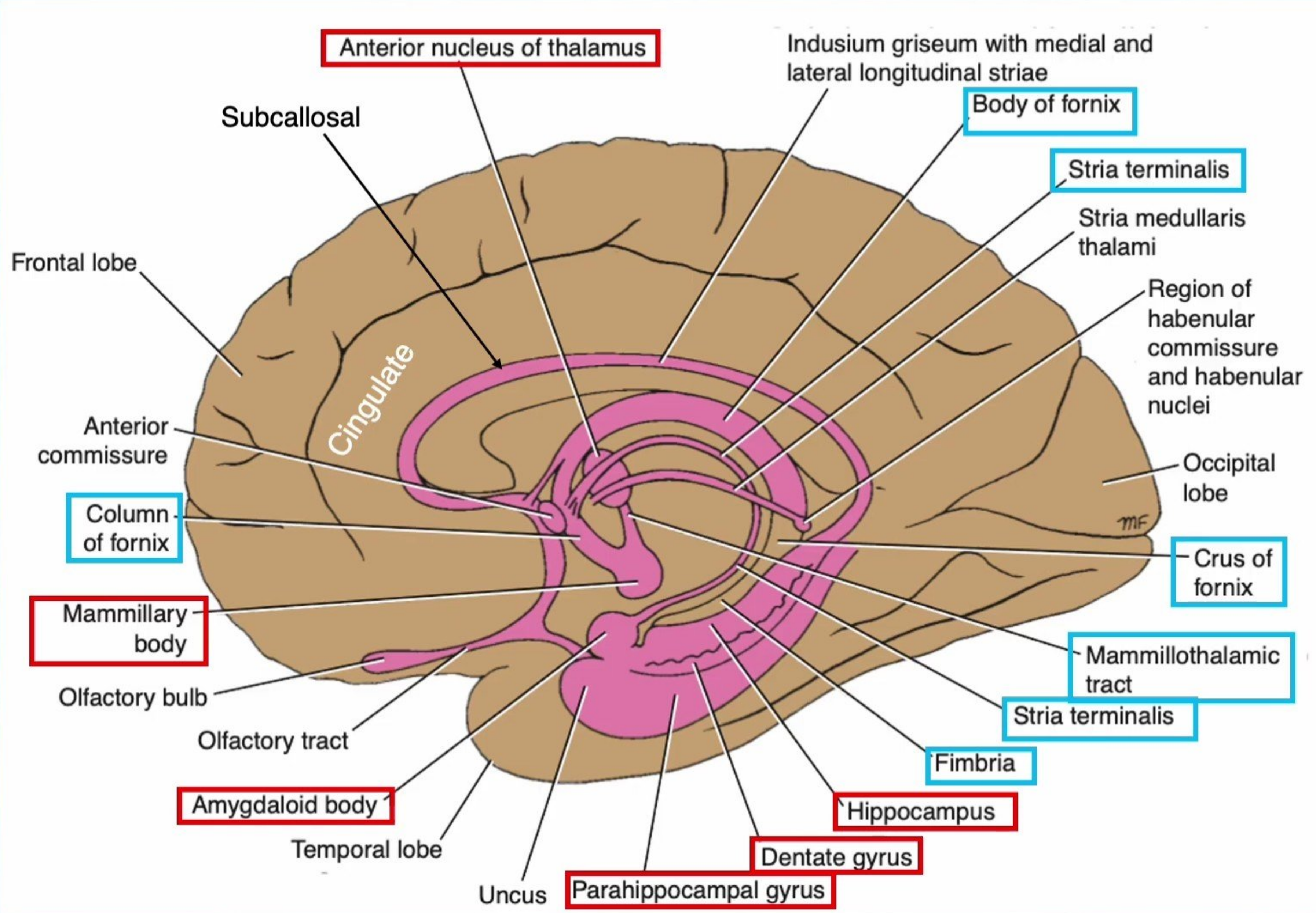
Limbic Lobe (The limbic system)

- The limbic system was loosely used to include a group of structures that lie in the border zone between the cerebral cortex and the hypothalamus.
- The limbic system is the control of emotion, behavior, drive,and important memory.

Anatomically, the limbic structures include:

- ✓The subcallosal.
- ✓The cingulate.
- ✓The parahippocampal gyri.
- ✓The hippocampal formation.
- ✓The amygdaloid nucleus.
- ✓The mammillary bodies.
- ✓The anterior thalamic nucleus.
- ✓The fimbria, the fornix, the mammillothalamic tract, and the stria terminalis constitute **the connecting pathways** of this system.

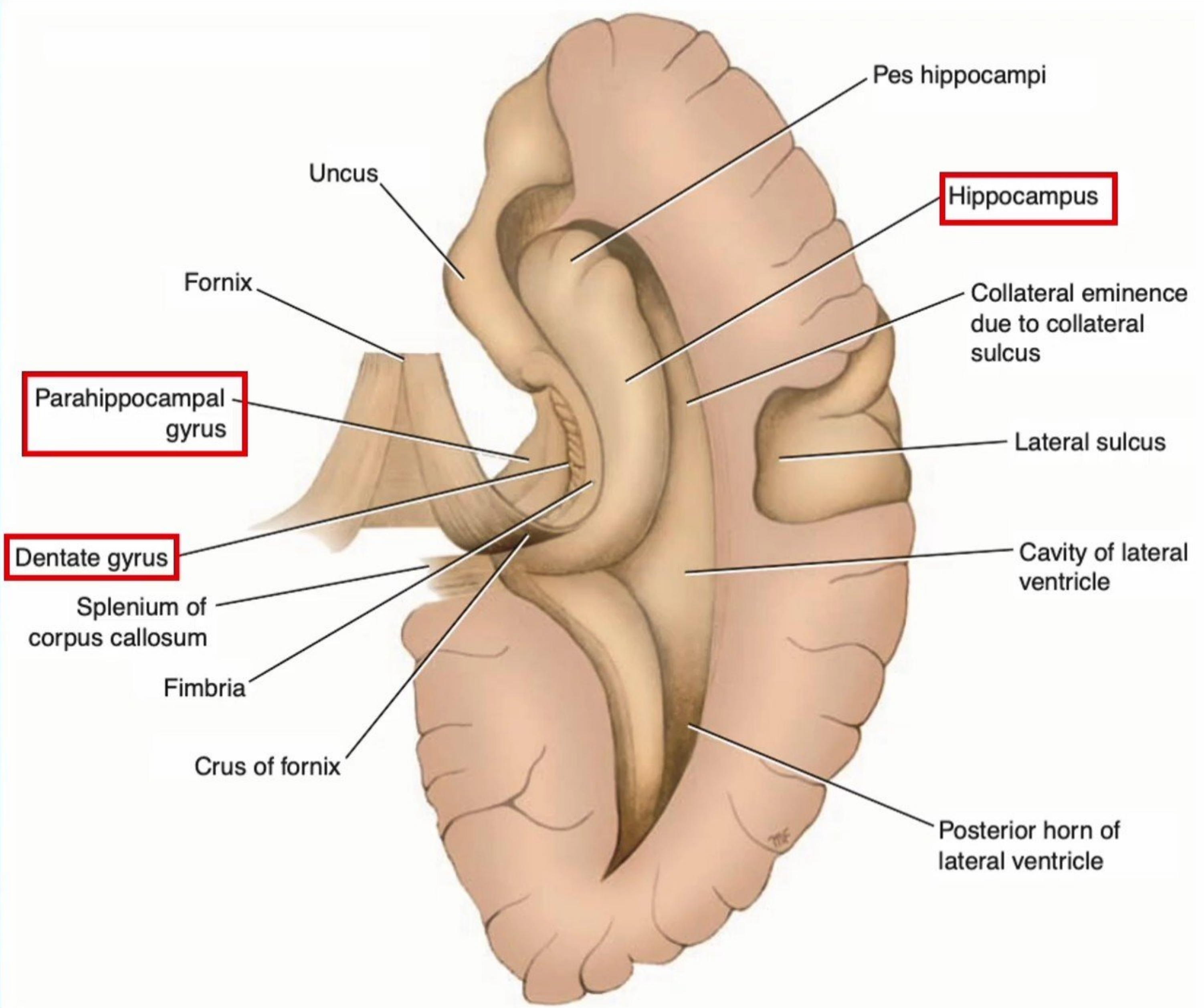
Limbic Lobe (The limbic system)



Limbic Lobe

The hippocampal formation consists of

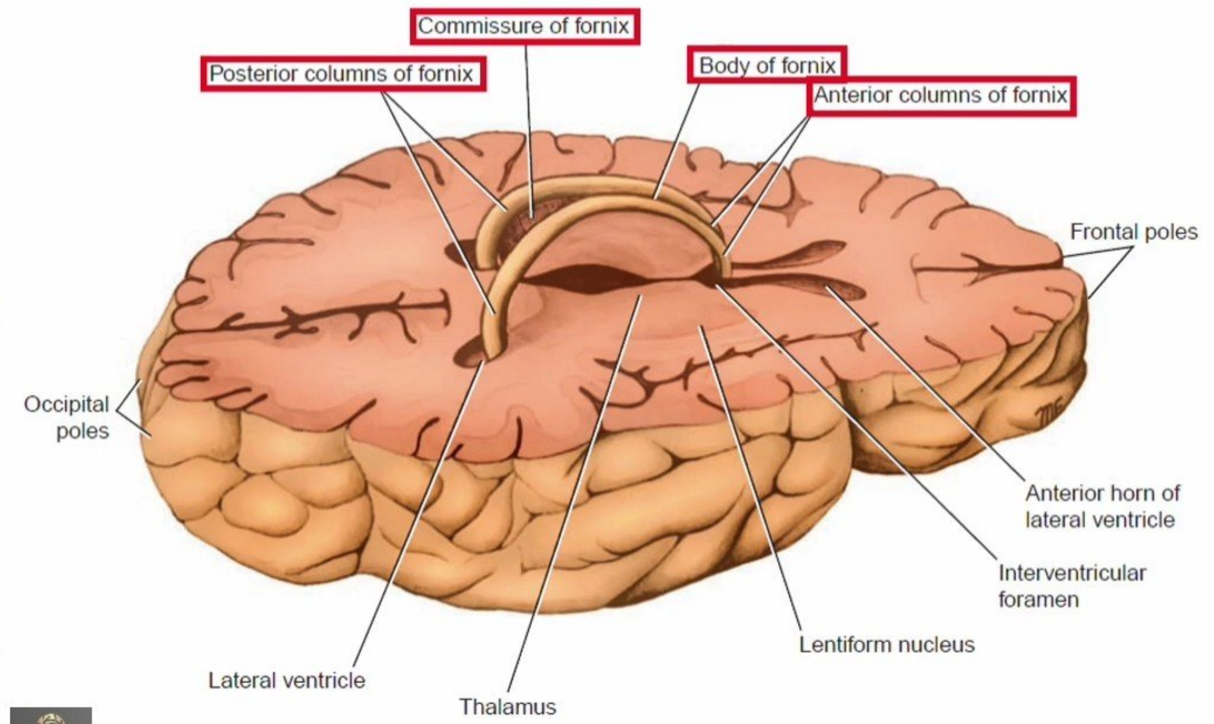
- ✓The hippocampus.
- ✓The dentate gyrus.
- ✓The parahippocampal gyrus.

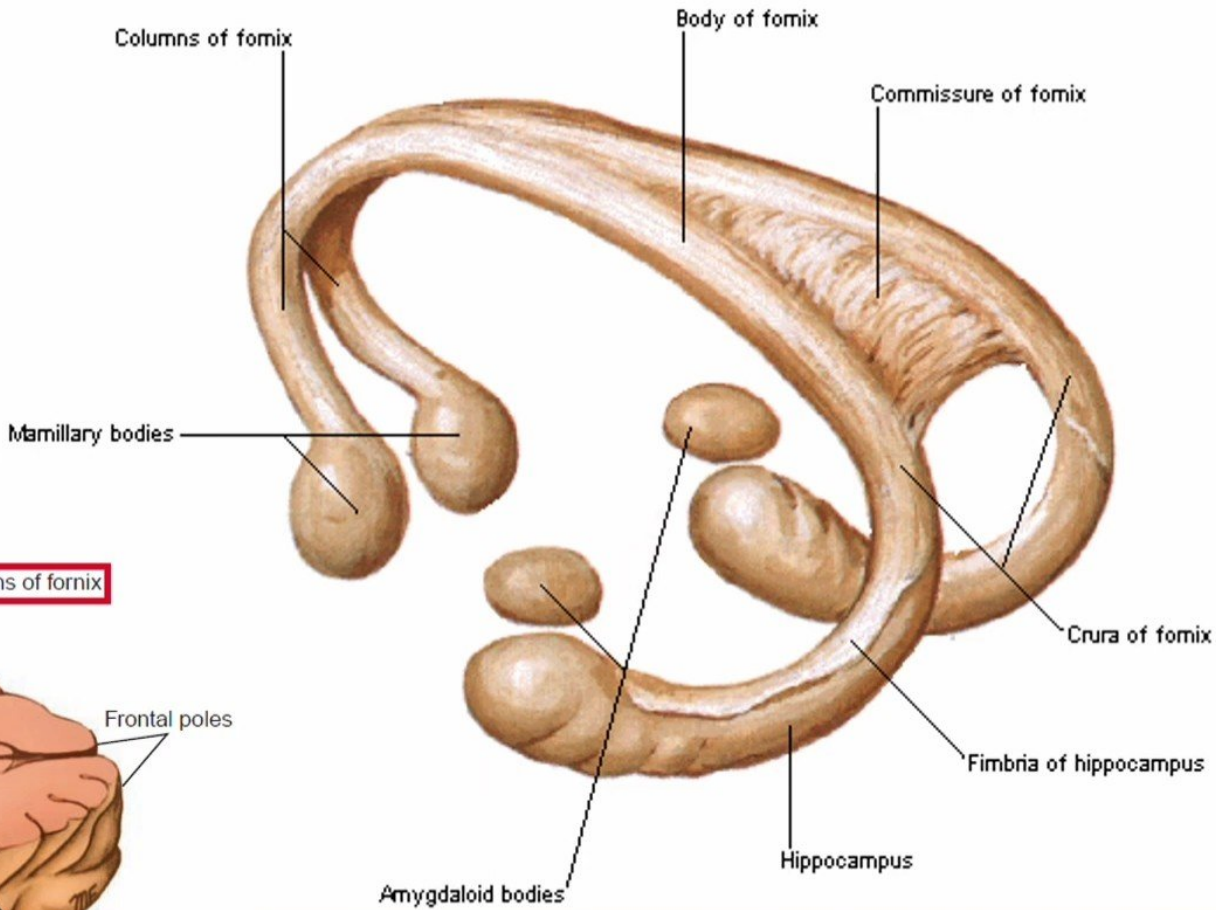


Limbic Lobe

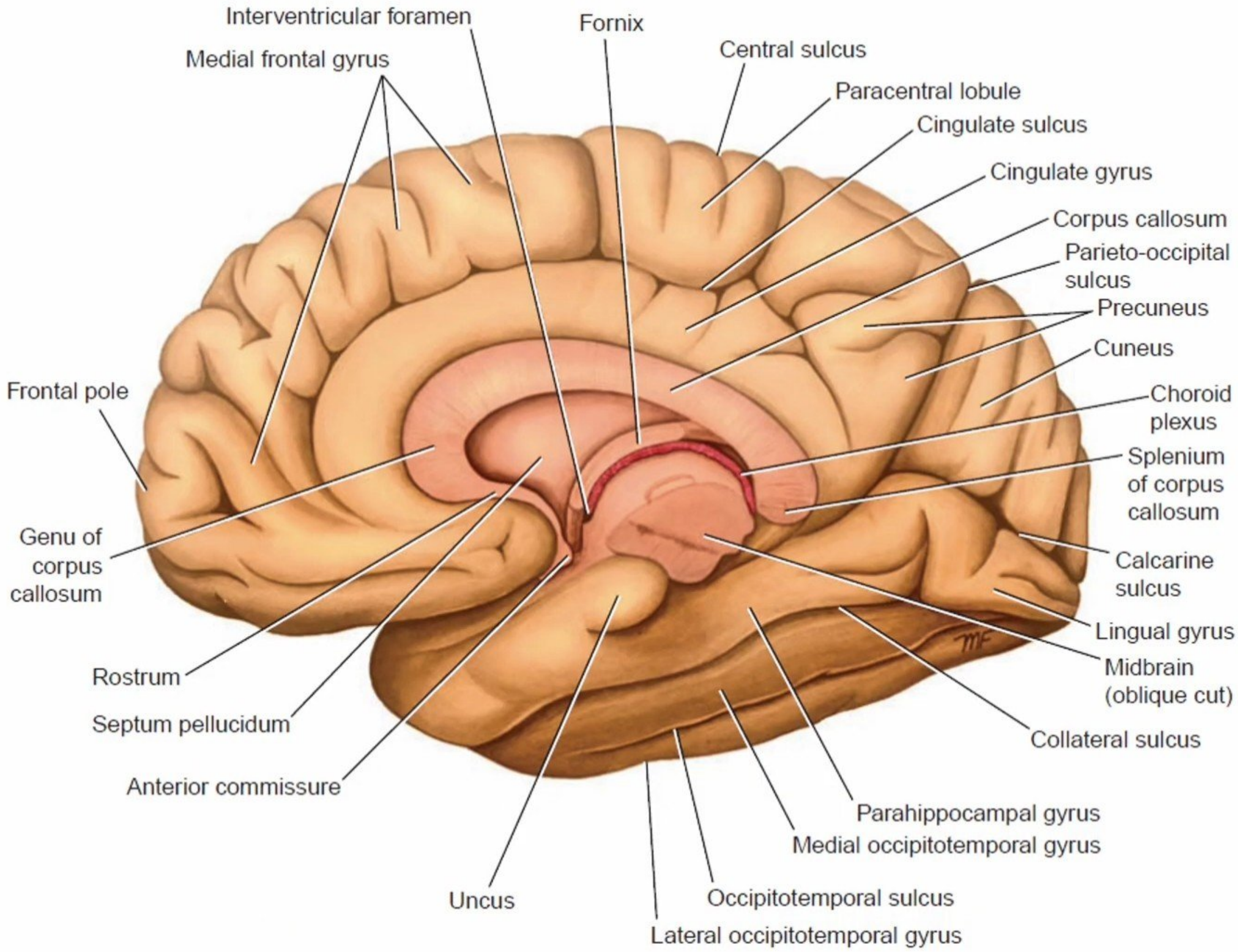
Consists of:

- ✓Hippocampal formation.
- ✓Amygdaloid nucleus.
- ✓Fornix.
- ✓Mammillary bodies.





Limbic Lobe

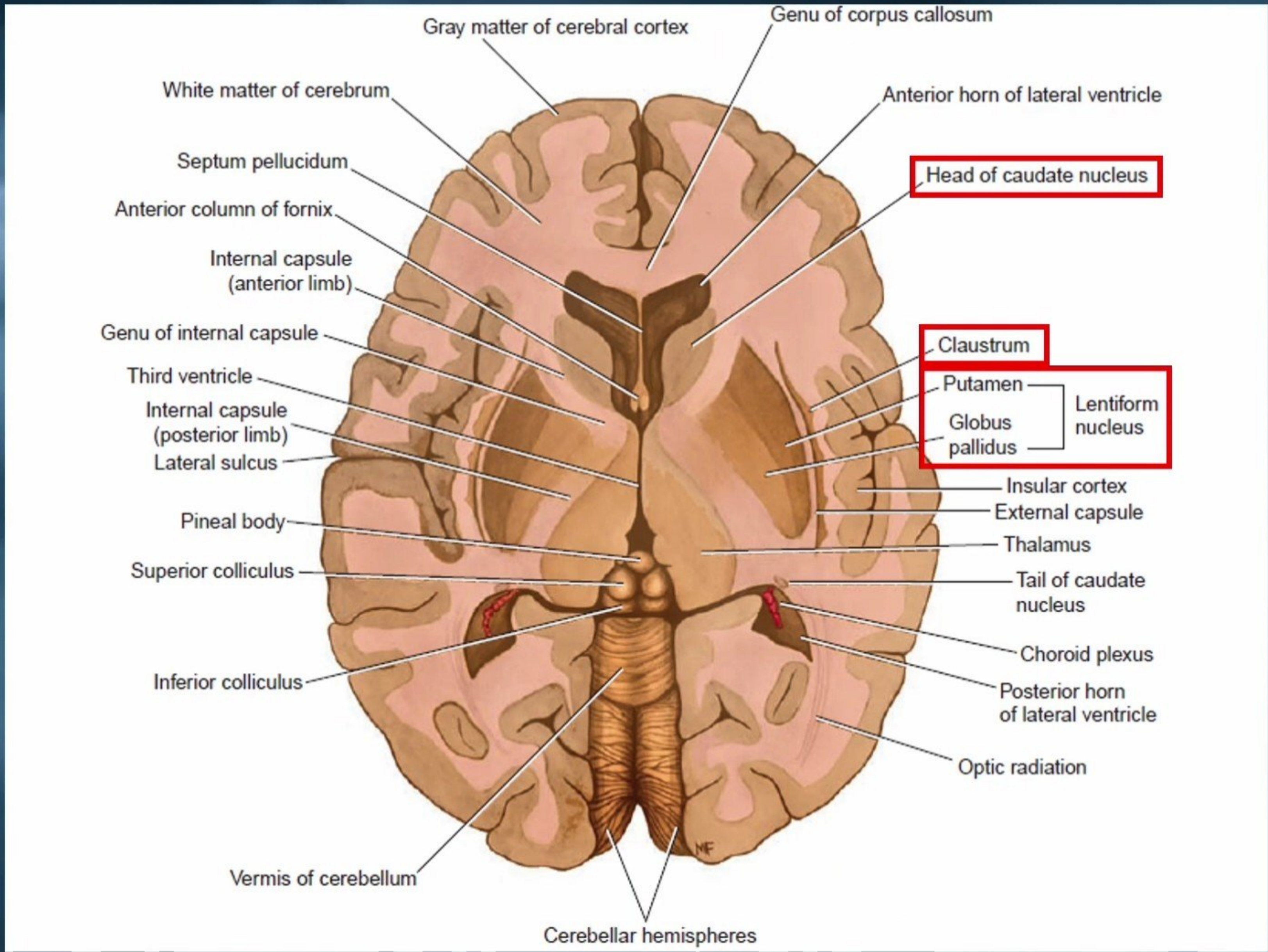


Interventricular foramen
Fornix
Medial frontal gyrus
Central sulcus
Paracentral lobule
Cingulate sulcus
Cingulate gyrus
Corpus callosum
Parieto-occipital sulcus
Precuneus
Cuneus
Frontal pole
Choroid plexus
Splenium of corpus callosum
Genu of corpus callosum
Calcarine sulcus
Lingual gyrus
Rostrum
Midbrain (oblique cut)
Septum pellucidum
Collateral sulcus
Anterior commissure
Parahippocampal gyrus
Medial occipitotemporal gyrus
Uncus
Occipitotemporal sulcus
Lateral occipitotemporal gyrus

Basal Nuclei

- ✓ The term basal nuclei (**basal ganglia**) is applied to a collection of masses of gray matter situated within each cerebral hemisphere.
- ✓ They are:

1) The **corpus striatum.**
   - ✓ It divided by a band of nerve fibers (**the**
   - ✓ **internal capsule**) into
   - ✓ The **caudate nucleus** and
   - ✓ **The lentiform nucleus**
2) The **amygdaloid** nucleus.
3) The **claustrum**.

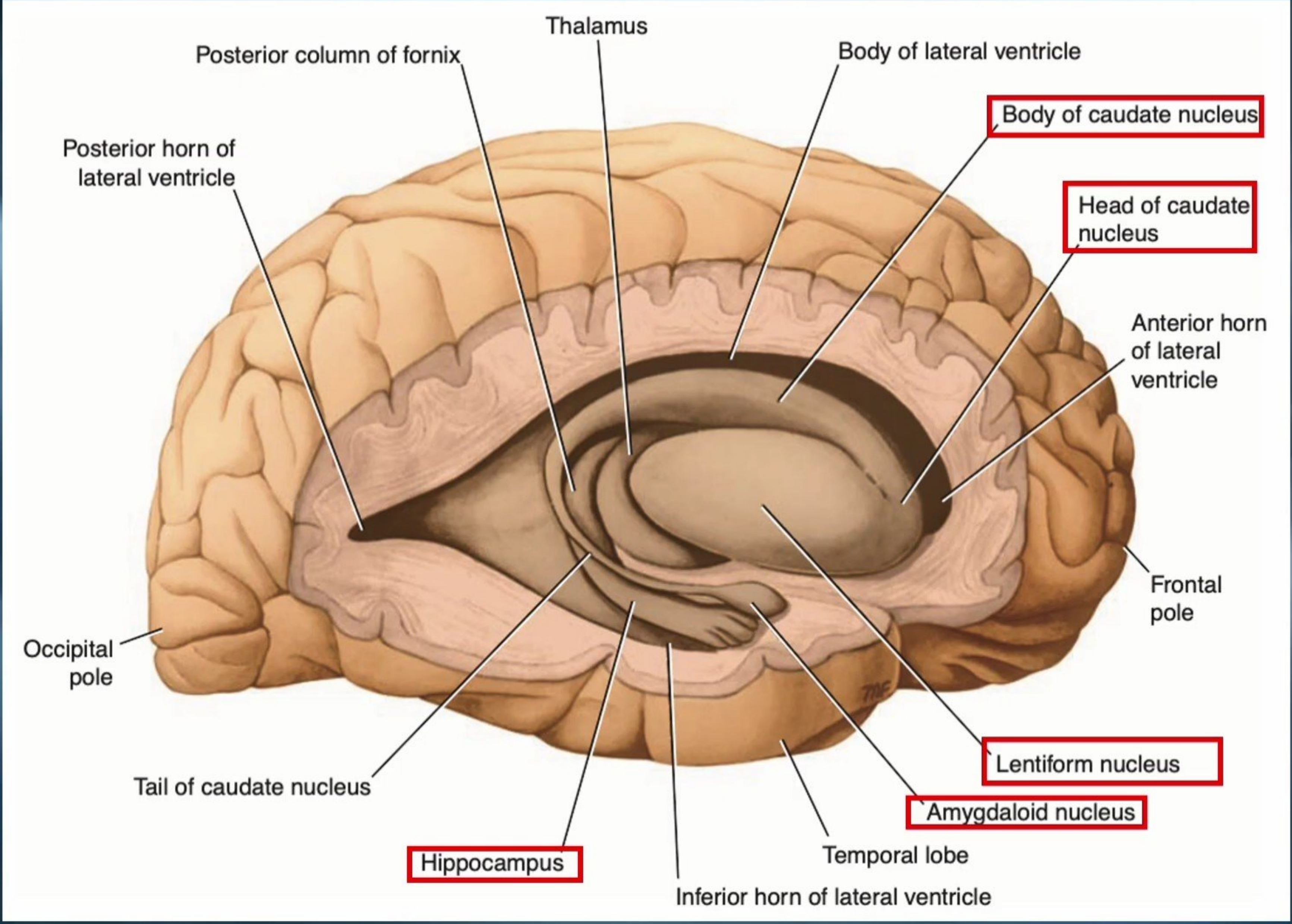


Lateral view of the right cerebral hemisphere dissected to show the position of:

- **The lentiform nucleus.**
- **The caudate nucleus.**
- **The thalamus.**
- **The hippocampus.**

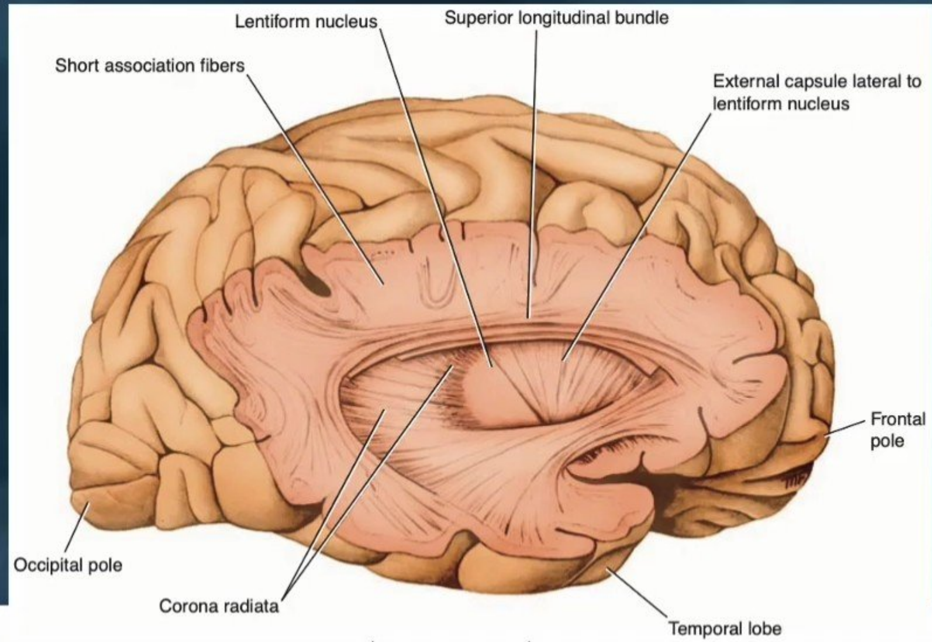
Basal ganglia:

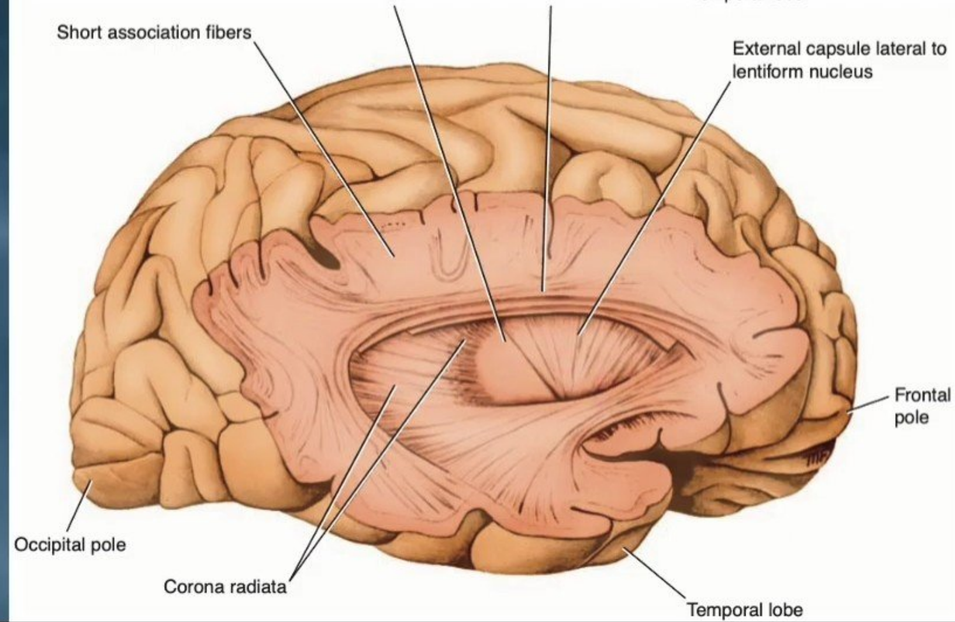
- ✓ It is control the intensity of movement and store learned movements.
- ✓ If damage, it causes Hemiballismus, Chorea, Rigidity and Tremor.



White Matter of the Cerebral Hemispheres

- The white matter is composed of myelinated nerve fibers of different diameters supported by neuroglia.
- The nerve fibers may be classified into three groups according to their connections:
  (1) Commissural fibers.
  (2) Association fibers.
  (3) Projection fibers.



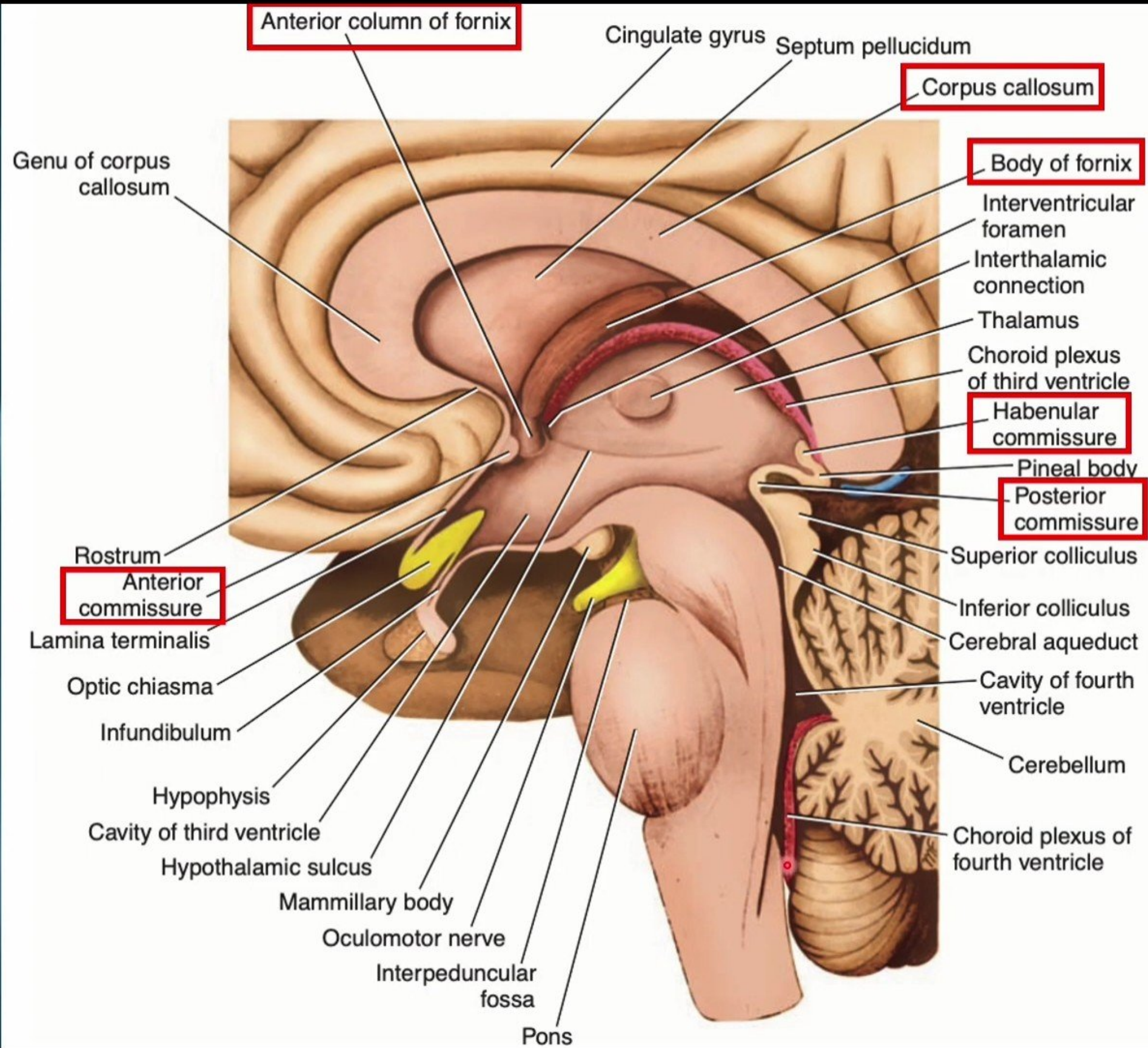


White Matter of the Cerebral Hemispheres

1) **Commissure fibers**
   They are **connected** corresponding regions of the **two hemispheres.**

➢ **They are as follows:**

1) The corpus callosum.
2) The anterior commissure.
3) The posterior commissure.
4) The fornix.
5) The habenular commissure.



White Matter of the Cerebral Hemispheres

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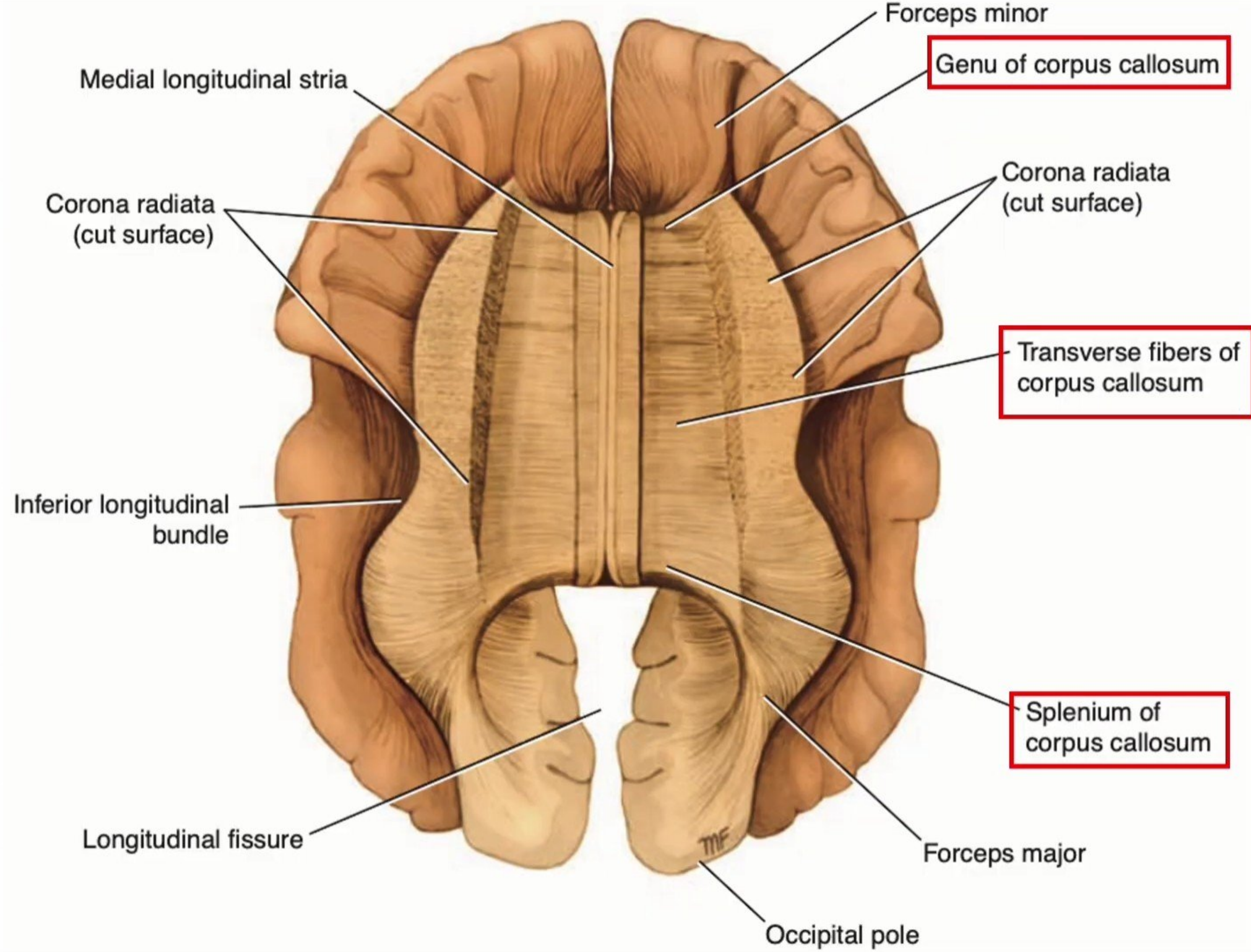
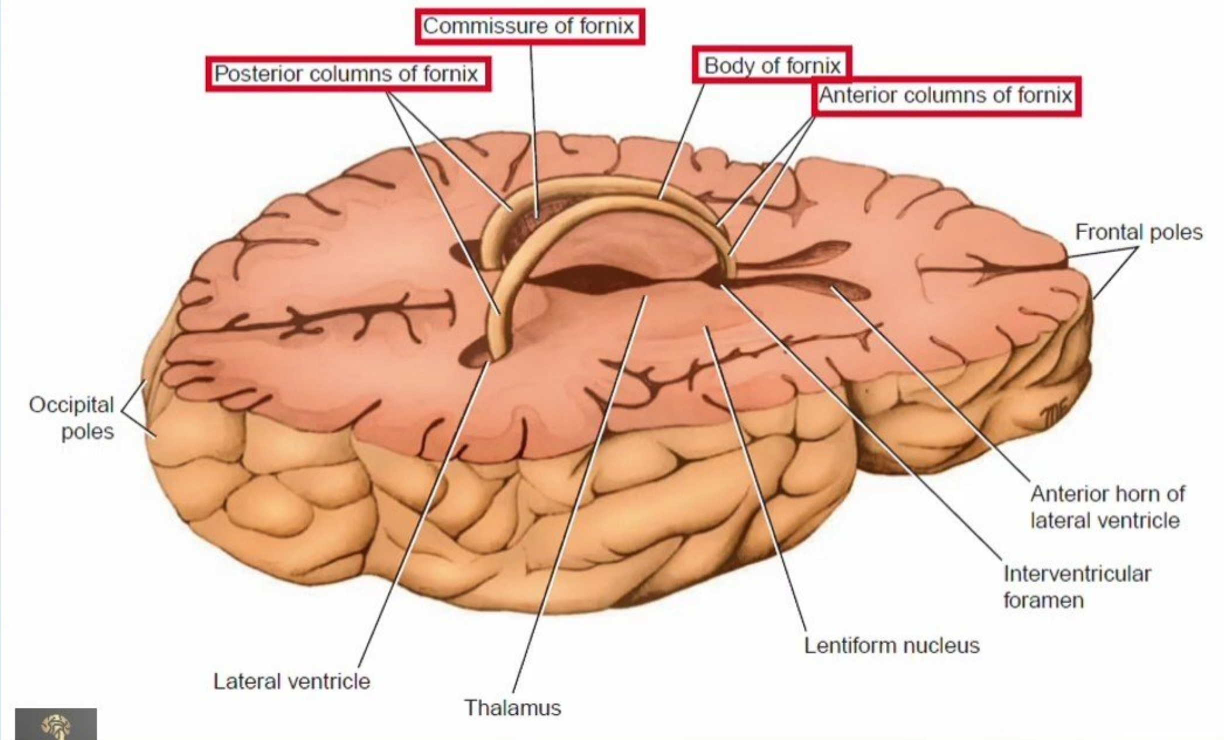
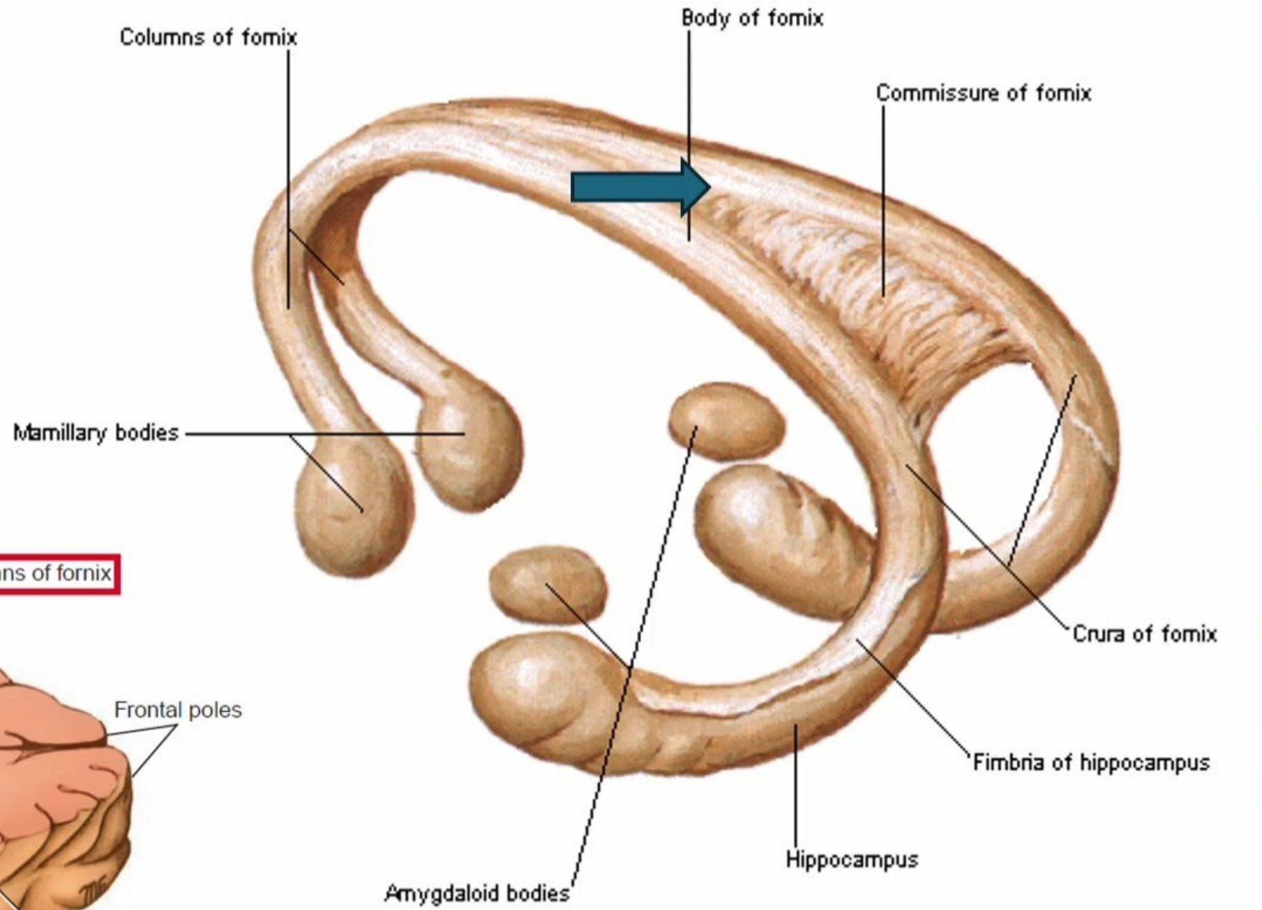


1) Commissure fibers

The fornix

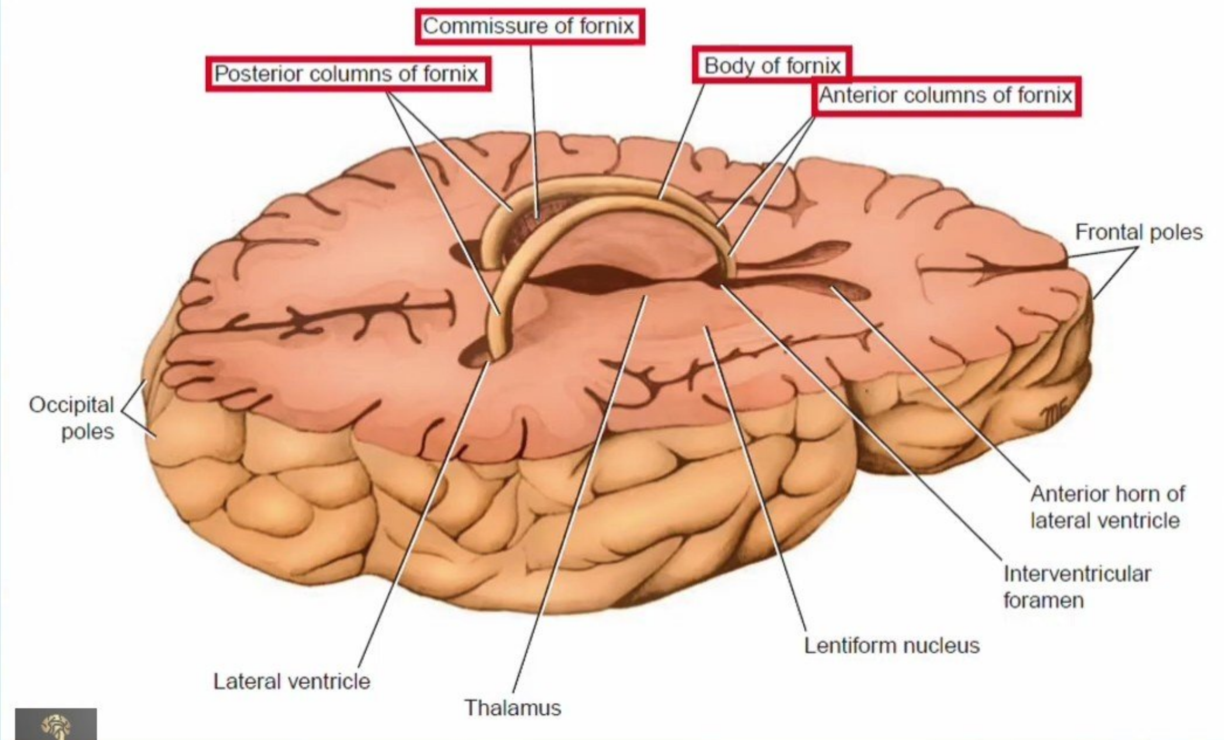


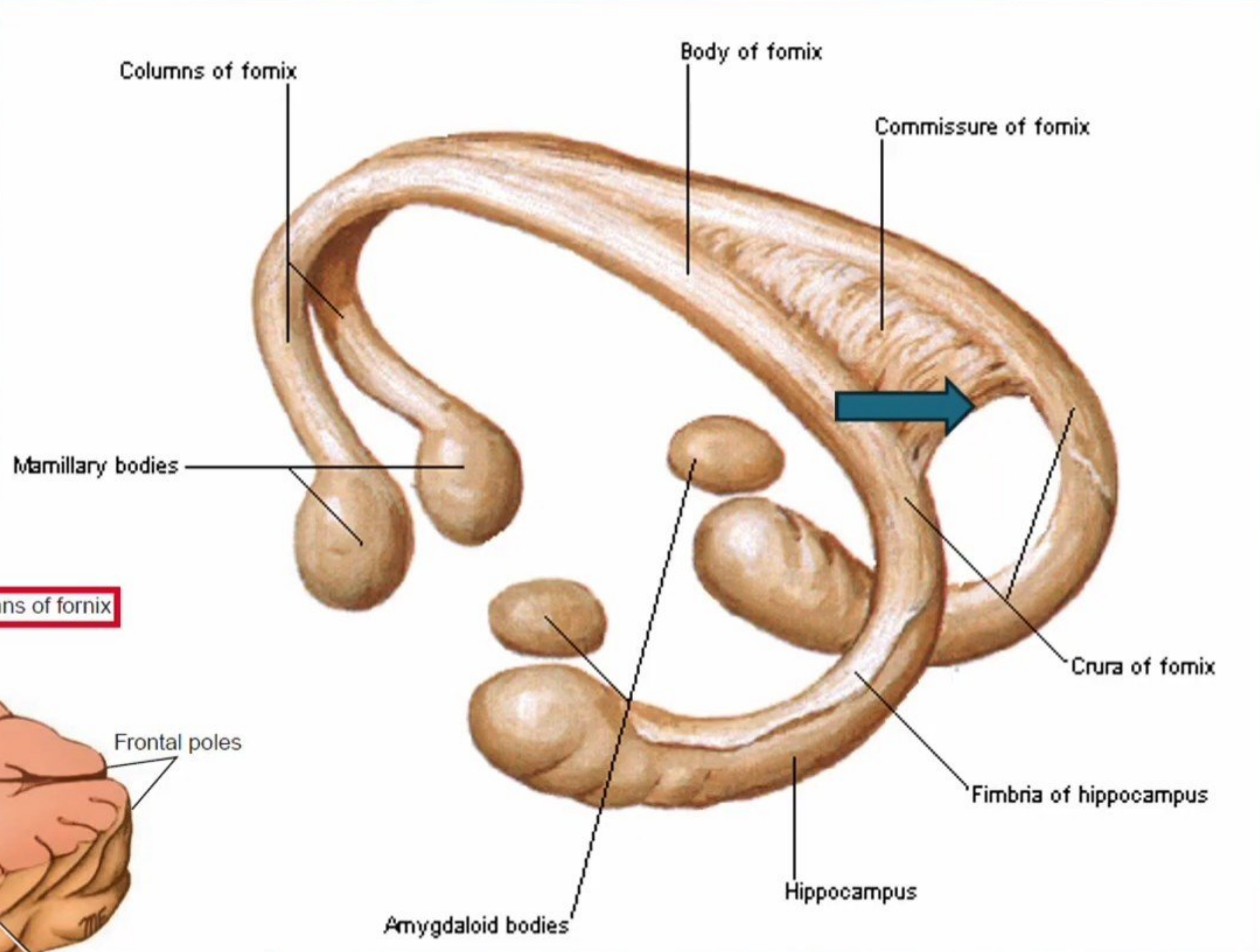


The function of the commissure of the fornix is to connect the hippocampal formations of the two sides.

1) Commissure fibers

The fornix





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White Matter of the Cerebral Hemispheres

2) Association fibers:

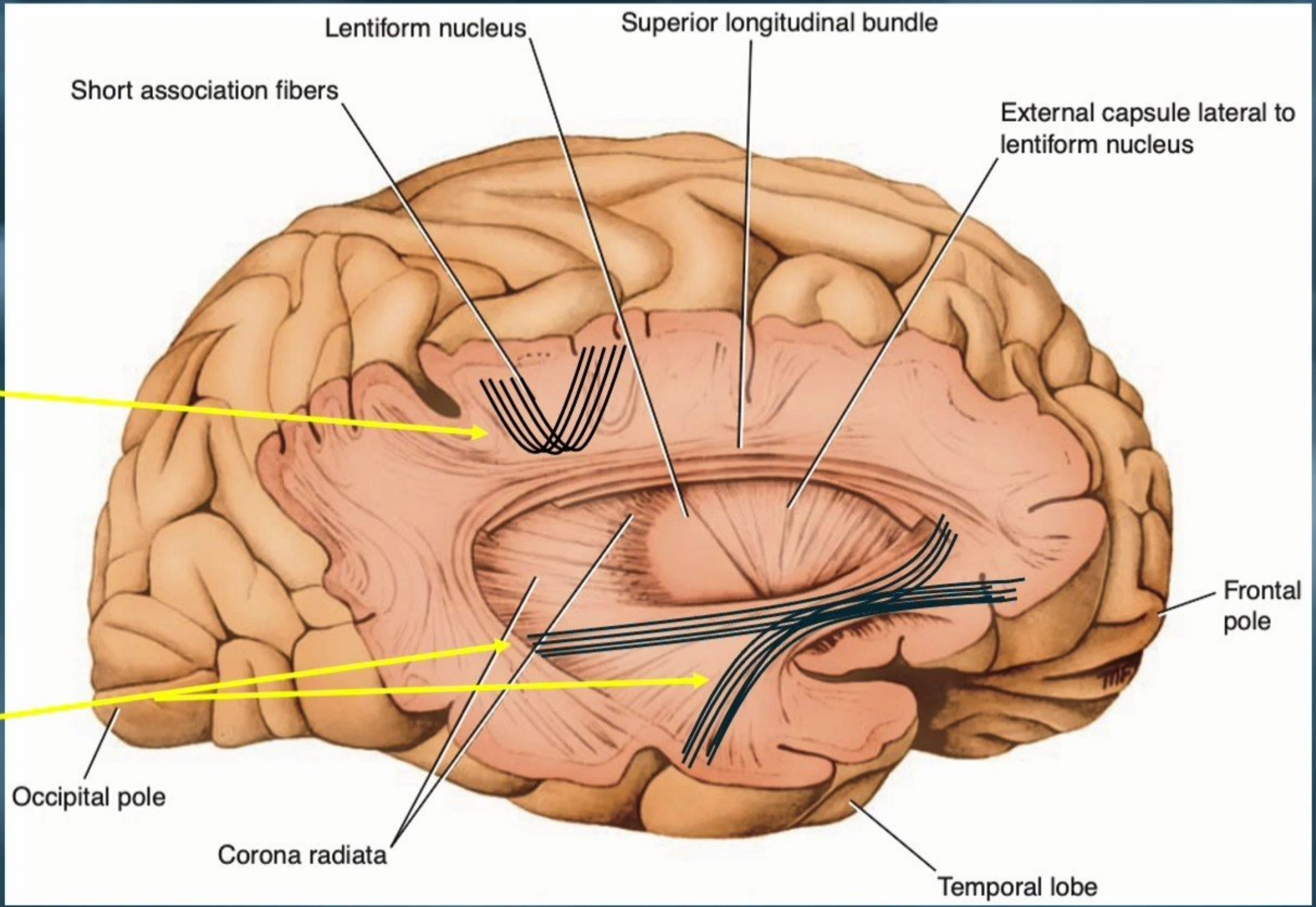
- They are nerve fibers that connect various cortical regions within the same hemisphere.
- It is divided into short and long groups.

A) The short association fibers :

- ✓ It lie immediately beneath the cortex and connect adjacent gyri.
- ✓ These fibers run transversely to the long axis of the sulci.

B) The long association fibers :

- ✓ They are collected into named bundles.



2) Association fibers:

B) The long association fibers :

The uncinate fasciculus:

- ✓ It connects the first motor speech area and the gyri on the inferior surface of the frontal lobe with the cortex of the pole of the temporal lobe.

The cingulum:

- ✓ It is curved fasciculus lying within the white matter of the cingulate gyrus.
- ✓ It connects the frontal and parietal lobes with parahippocampal and adjacent temporal cortical regions.

The superior longitudinal fasciculus:

- ✓ It is the largest bundle of nerve fibers.
- ✓ It connects the anterior part of the frontal lobe to the occipital and temporal lobes.

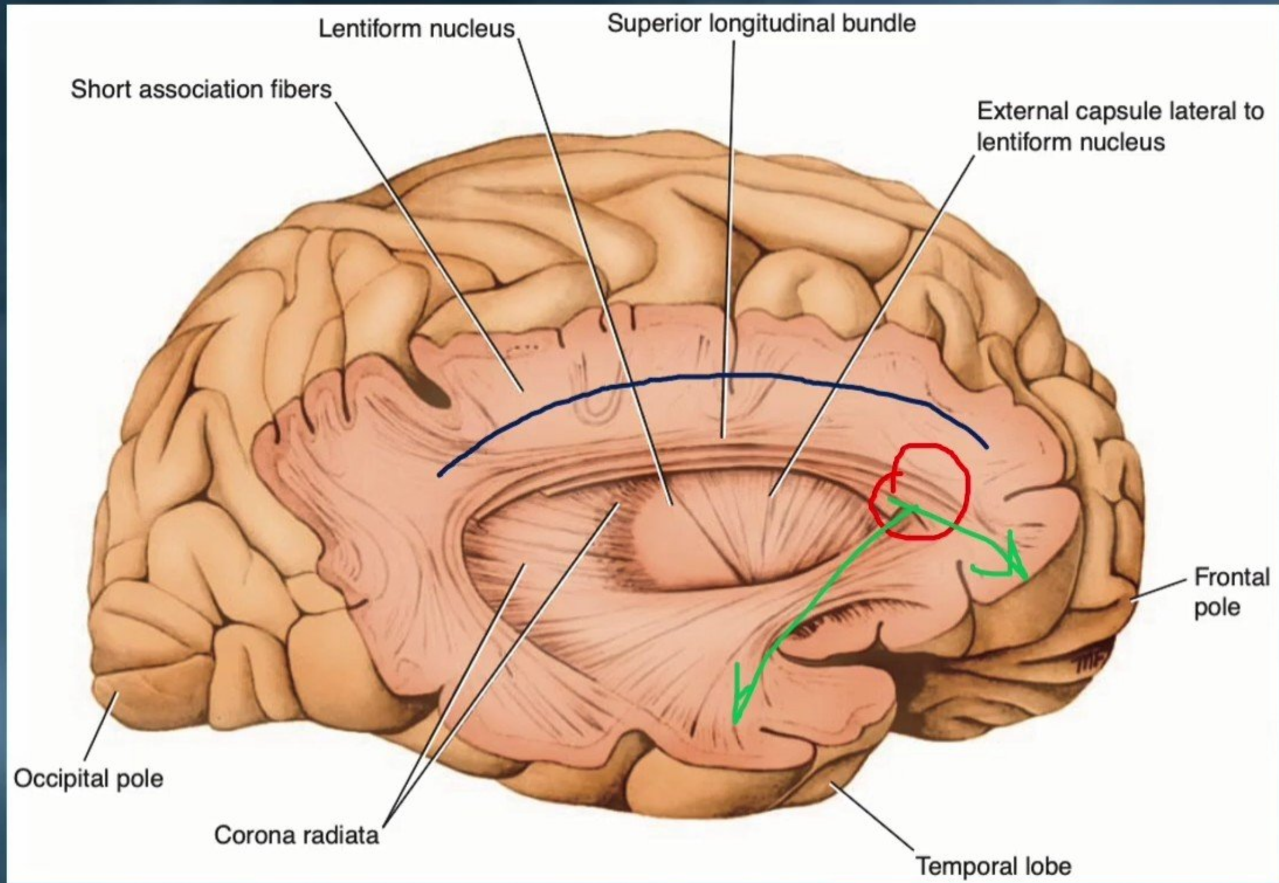
The inferior longitudinal fasciculus

- ✓ It runs anteriorly from the occipital lobe, passing lateral to the optic radiation, and to the temporal lobe.

The fronto-occipital fasciculus

- ✓ It connects the frontal lobe to the occipital and temporal lobes.

White Matter of the Cerebral Hemispheres



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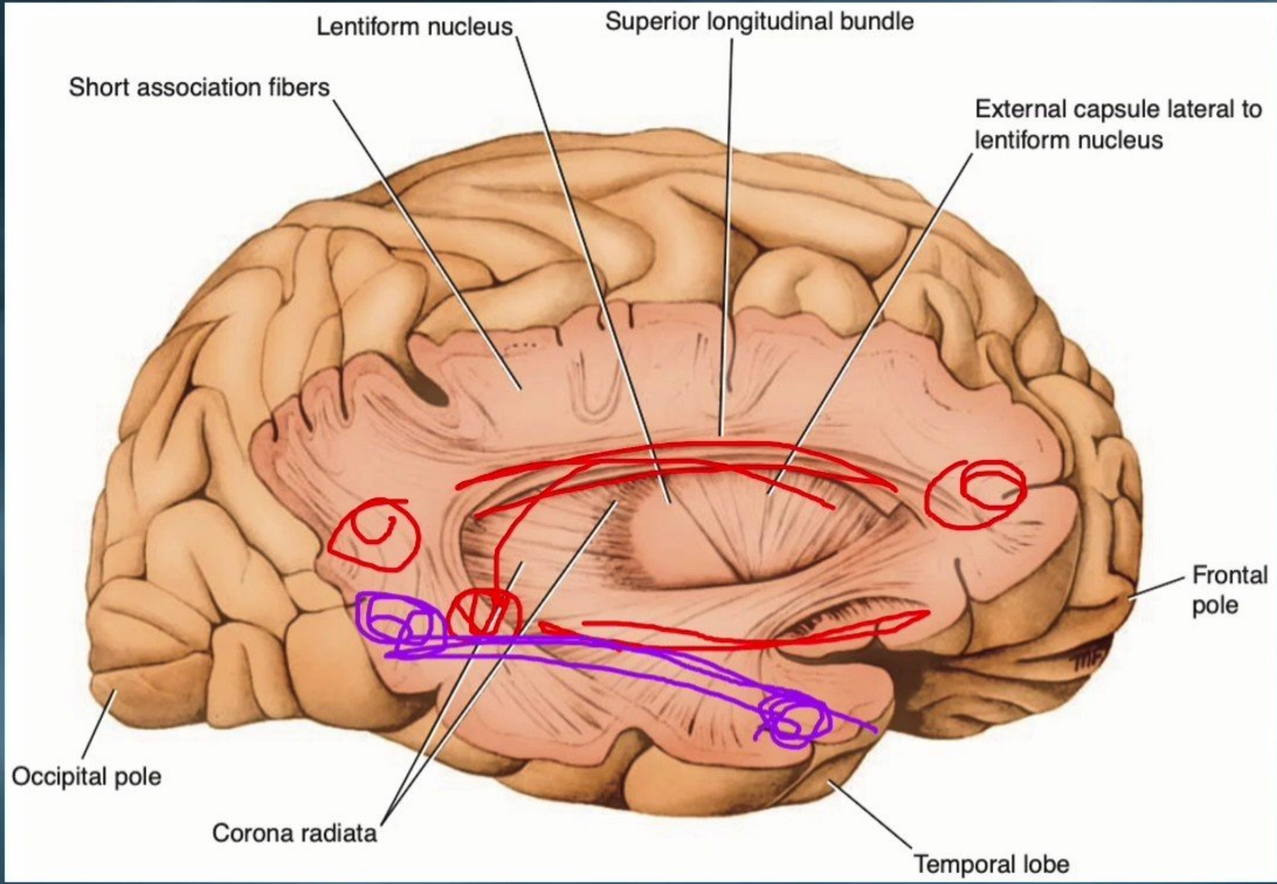

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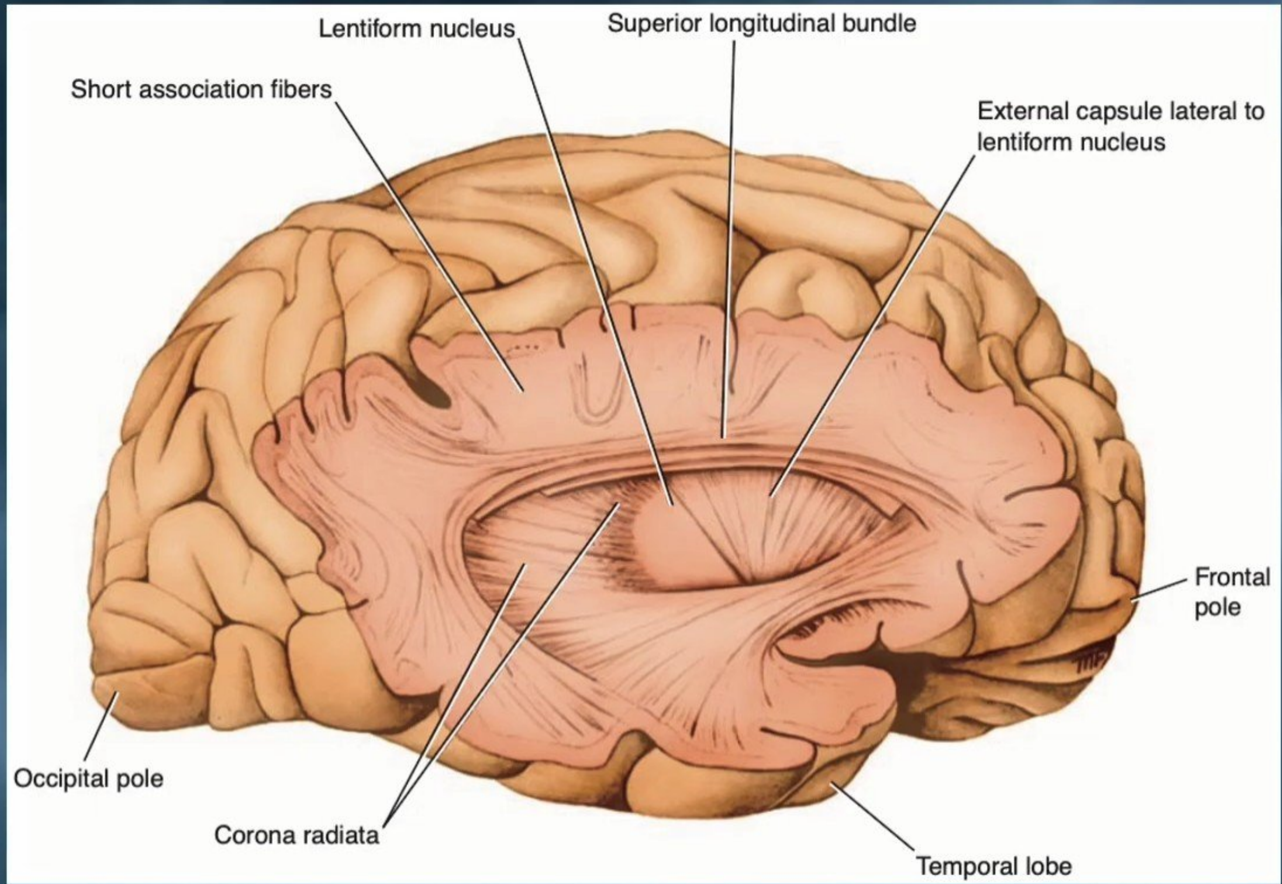
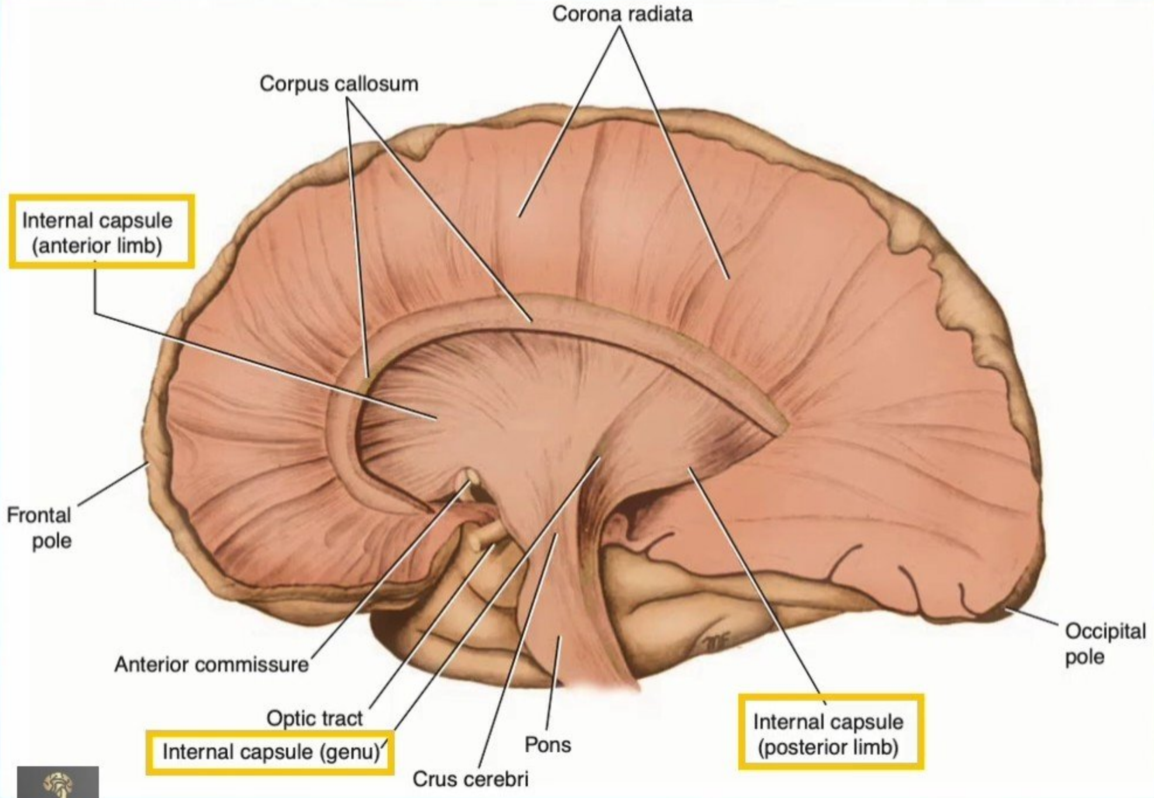
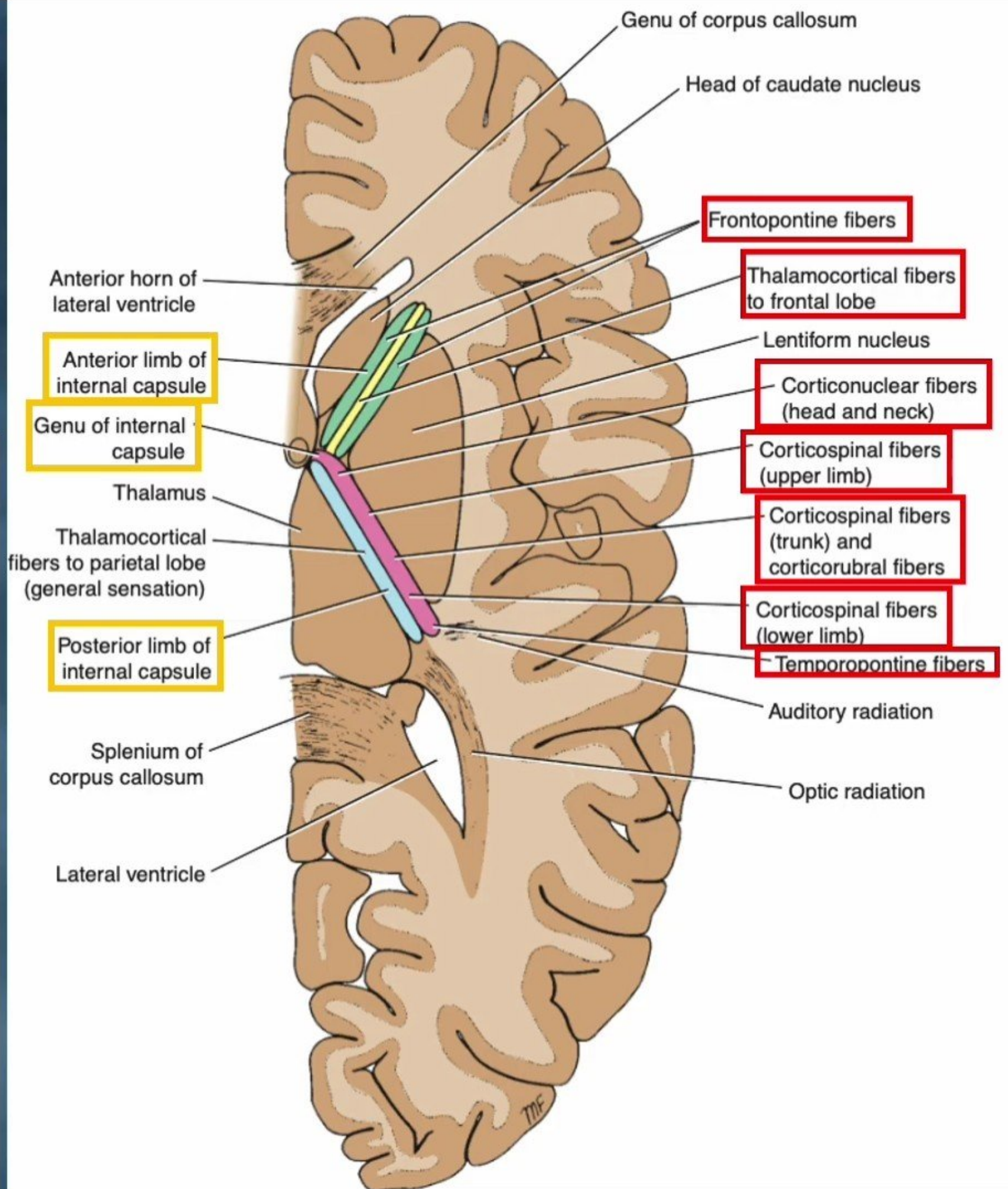


White Matter of the Cerebral Hemispheres

3) Projection Fibers:

- Afferent and efferent nerve fibers passing **to** and **from** the **brainstem** to the entire **cerebral cortex** must travel between large nuclear masses of gray matter within the cerebral hemisphere, These fibers form a compact band known as **internal capsule** and **corona radiata**.





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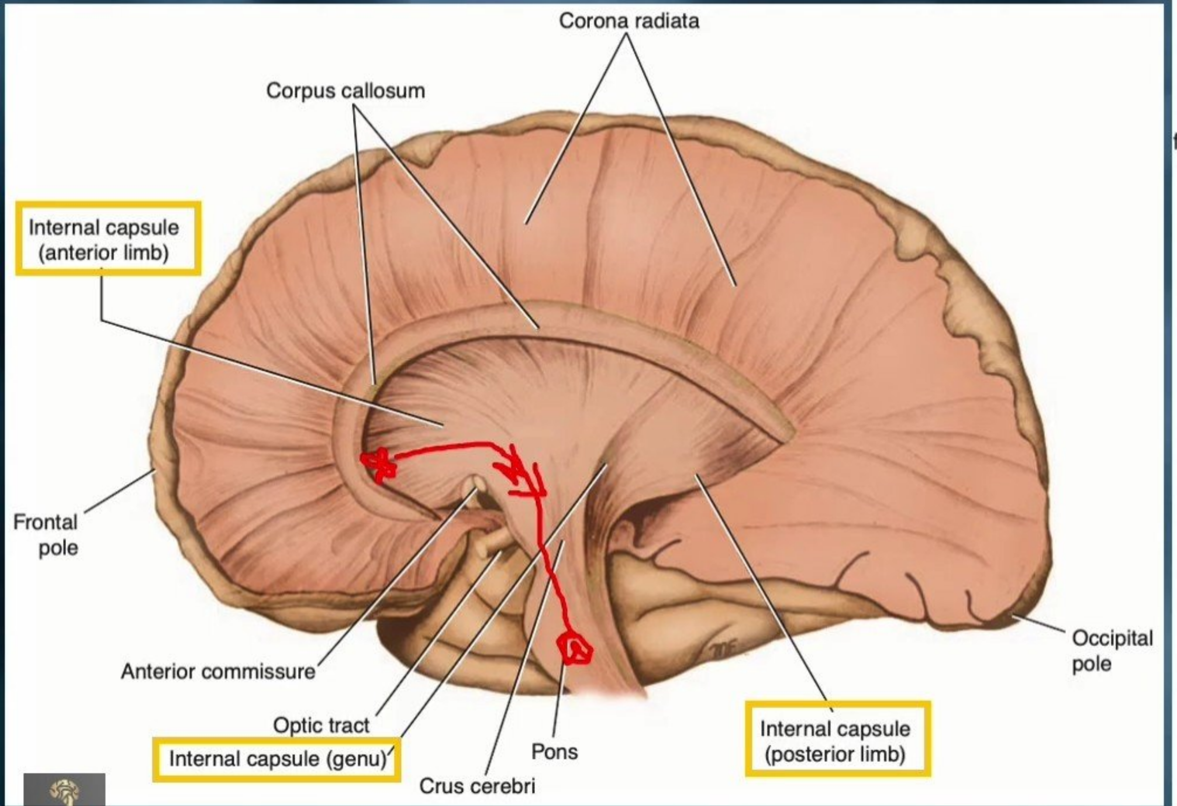
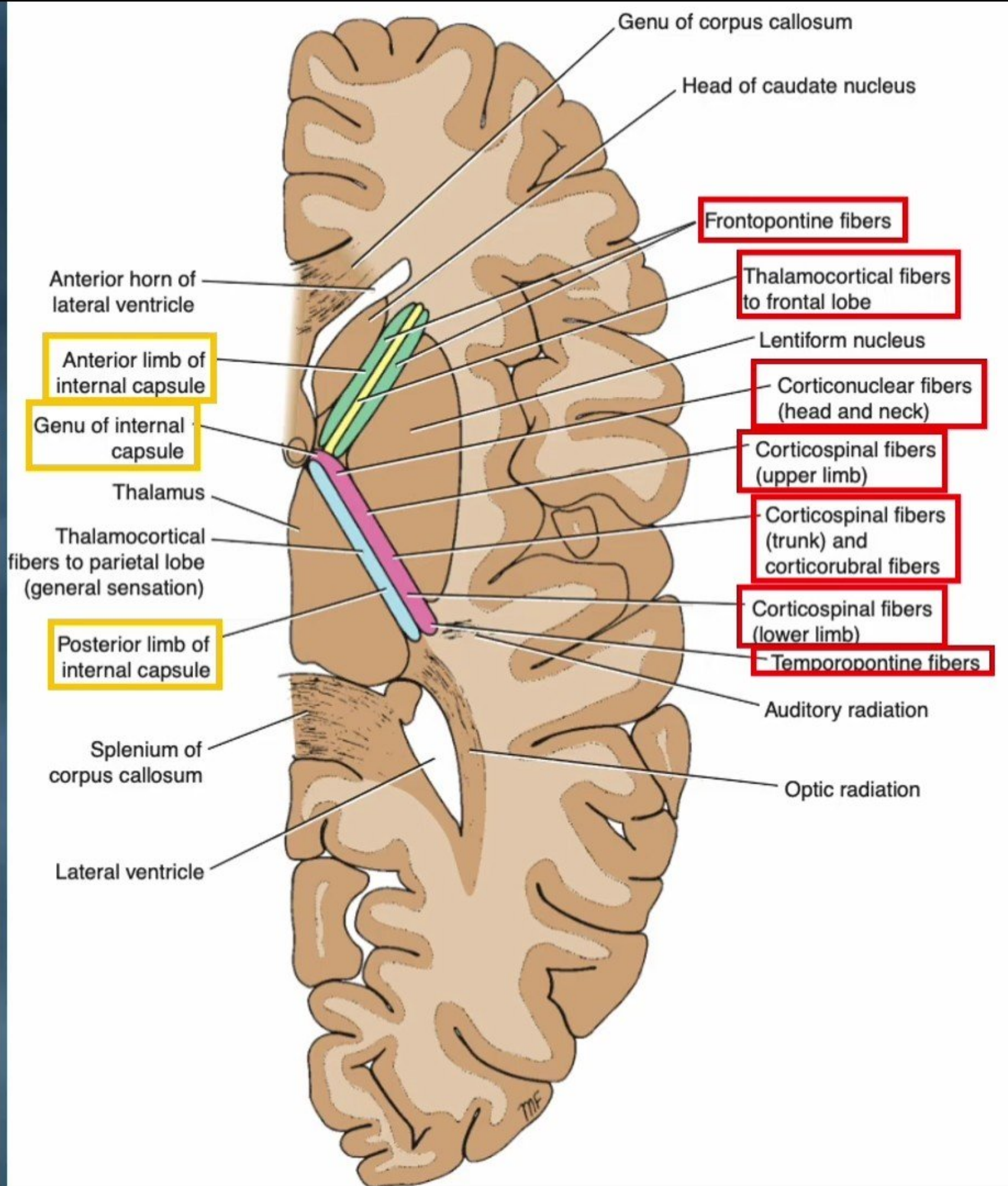
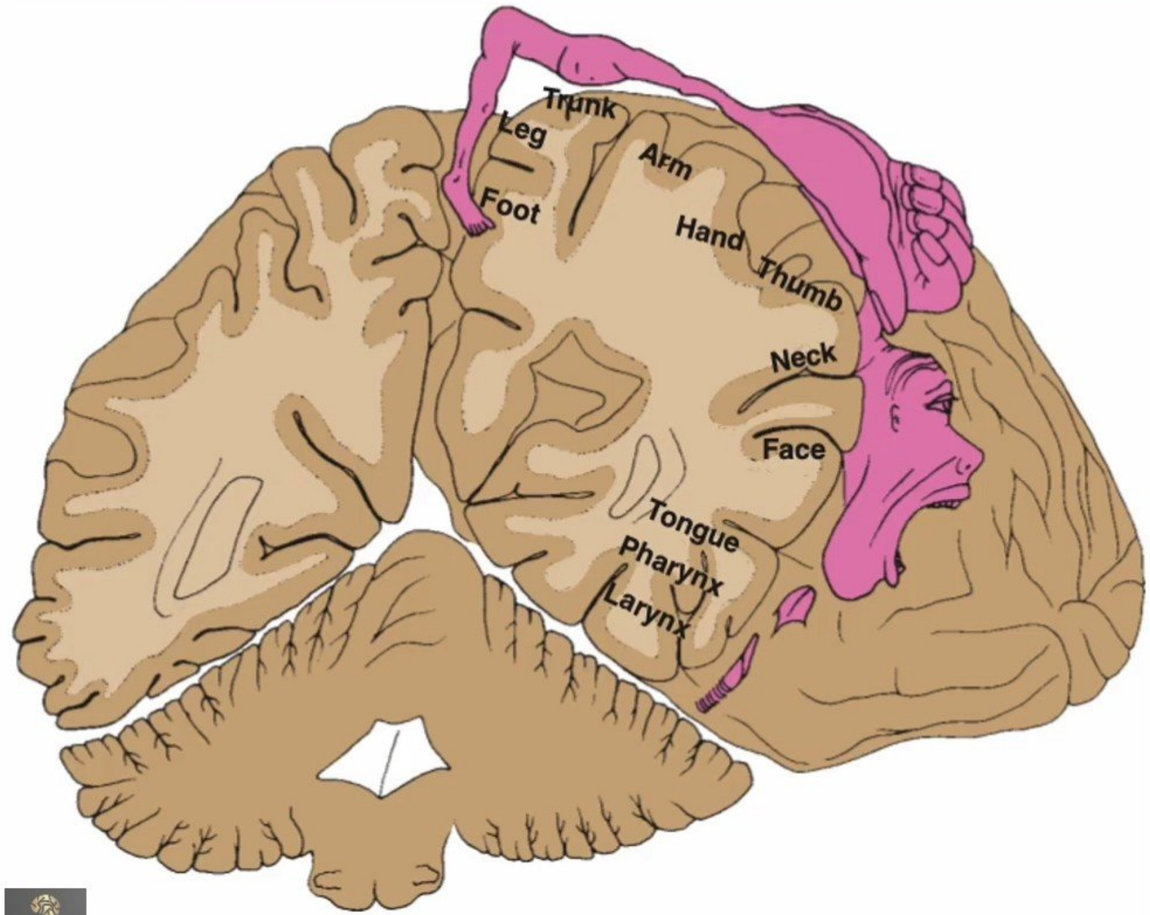




Lobes of Cerebral Cortex and Associated Integration

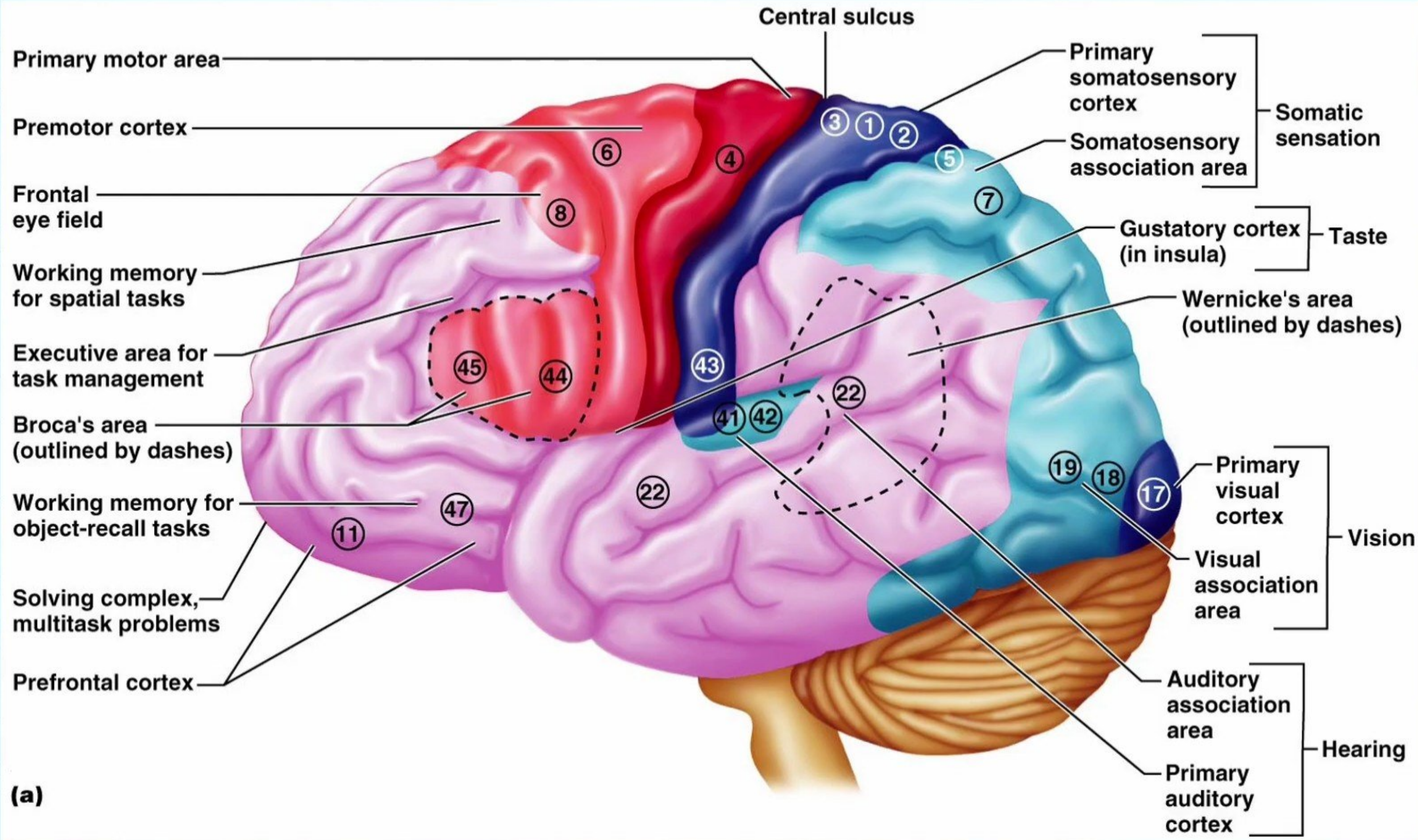
- **Frontal** – voluntary movement, behavior, perception.
- **Parietal** – tactile sensory.
- **Occipital** – vision.
- **Temporal** – olfactory, auditory & gustatory.
- **Limbic** – emotion, behavior, memory, personality.

Motor homunculus on the precentral gyrus



Functional Areas of the Cerebral Cortex

Brodmann's area

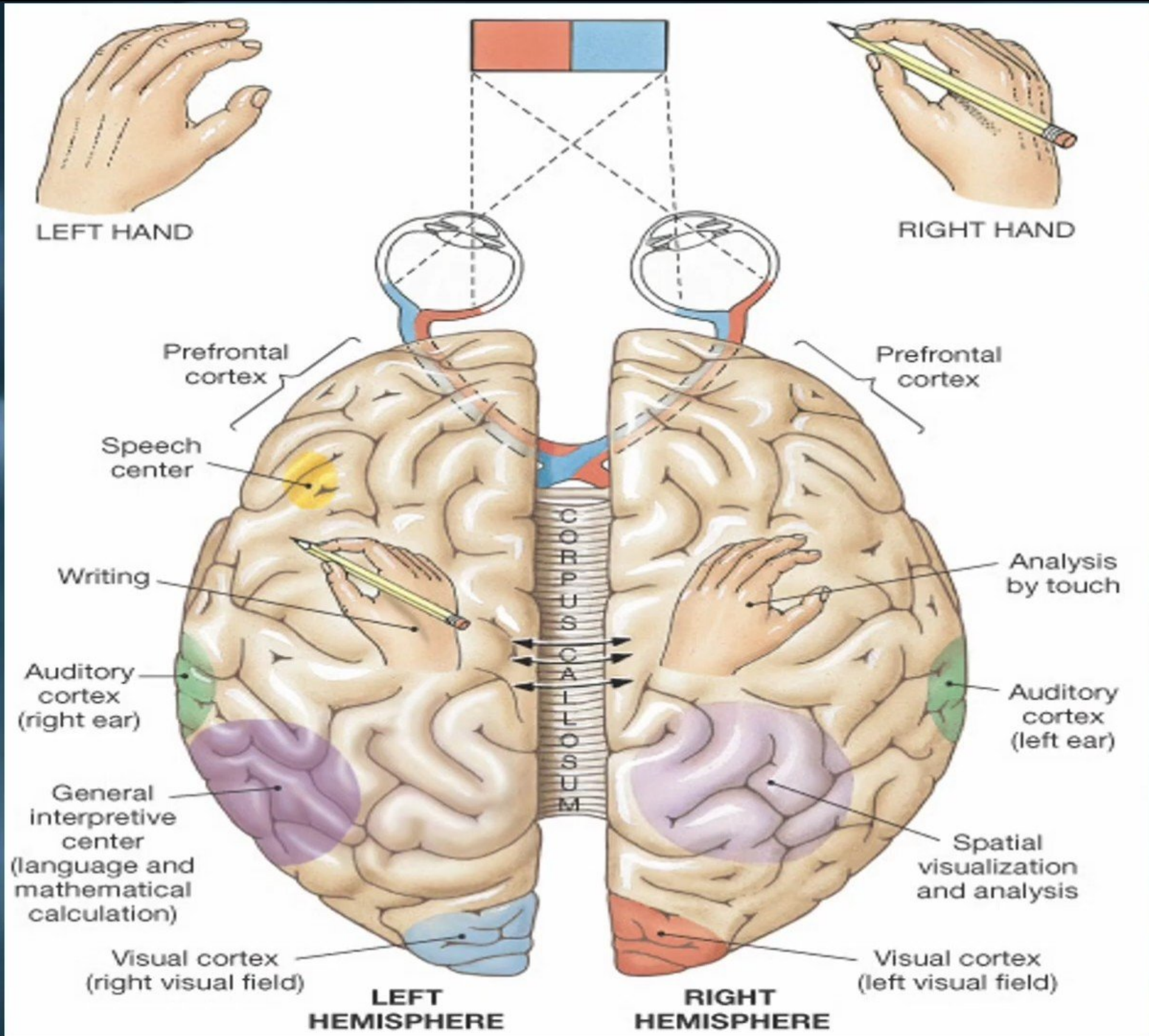


Cerebral Dominance

Consciousness

- ✓ **Right-handed individuals**

➢ **The left cerebral hemi-sphere is dominant**

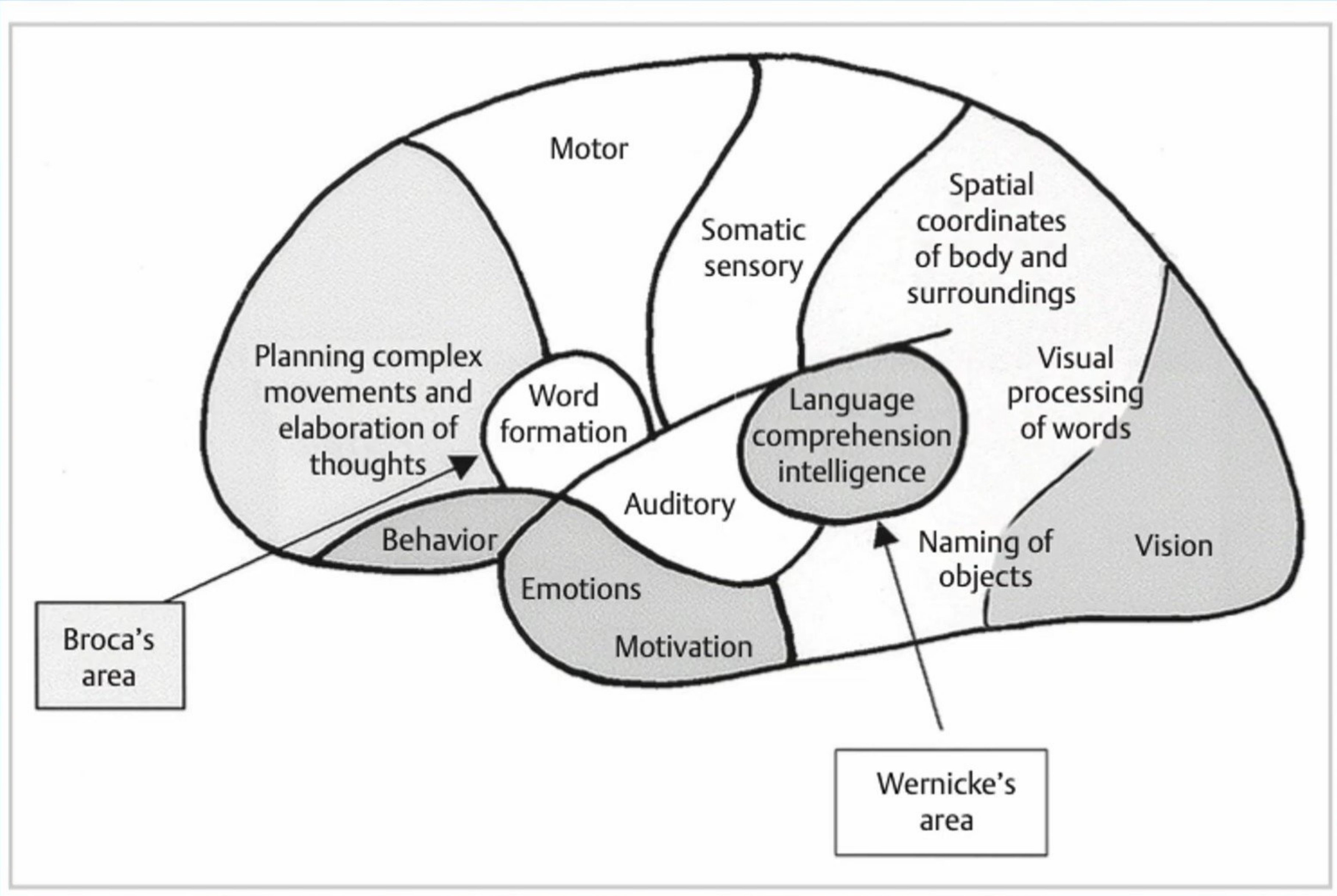


what Cortical Region of the brain would these doctors be stimulating?

"Whoa! *That* was a good one! Try it, Hobbs—just poke his brain right where my finger is."

Functional cortical areas

Association areas



Association areas

- It is distinct from the primary and secondary motor and sensory areas.
- They receive and analyze impulses from **several different cortical** and **subcortical regions**.

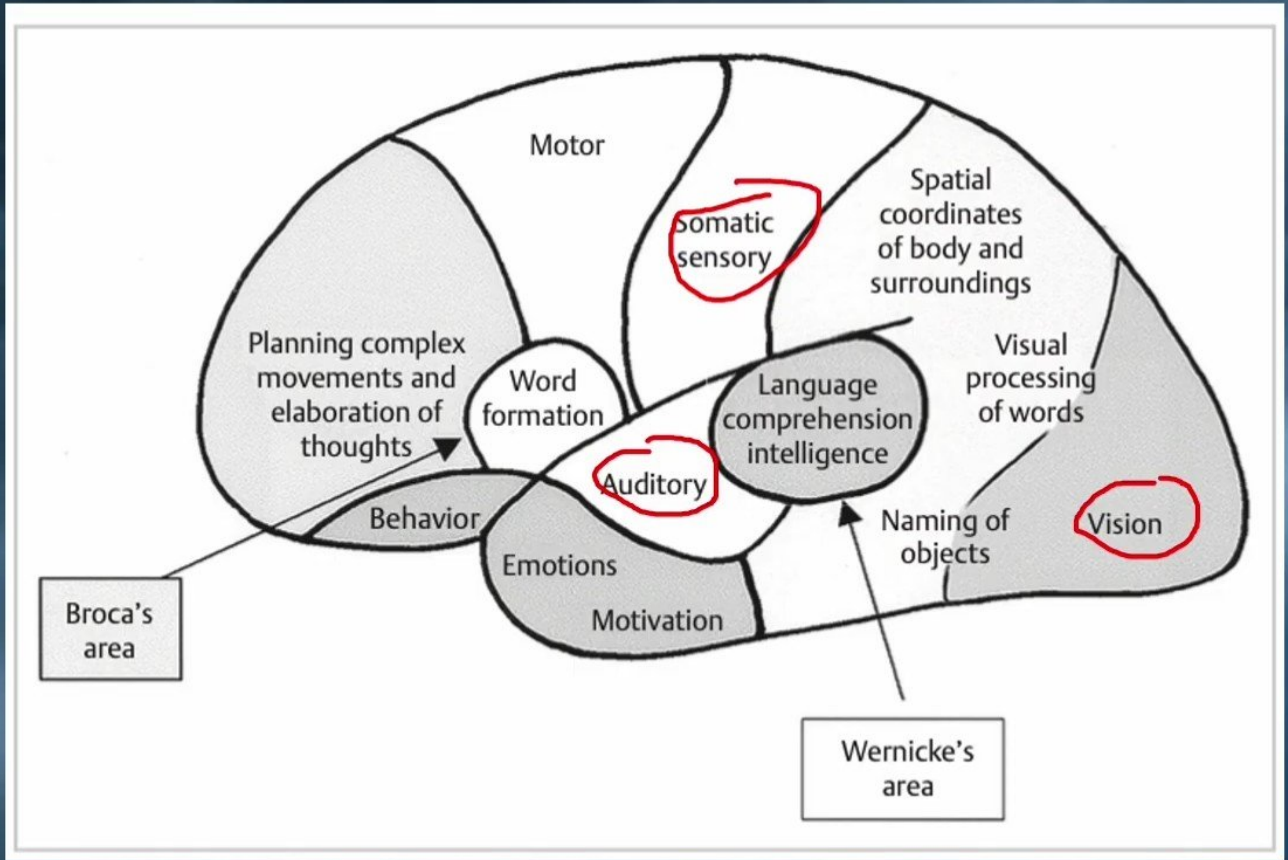
A) Parieto-occipito-temporal association cortex:

- It is located between the somatosensory, visual, and auditory cortices.
- It determines the spatial coordinates of the body and its surroundings.
- Damage of this area leads to neglect of the opposite side of the body/surroundings.

1) Wernicke's area.

2) Angular gyrus.

3) Nondominant parieto-temporo-occipital cortex.



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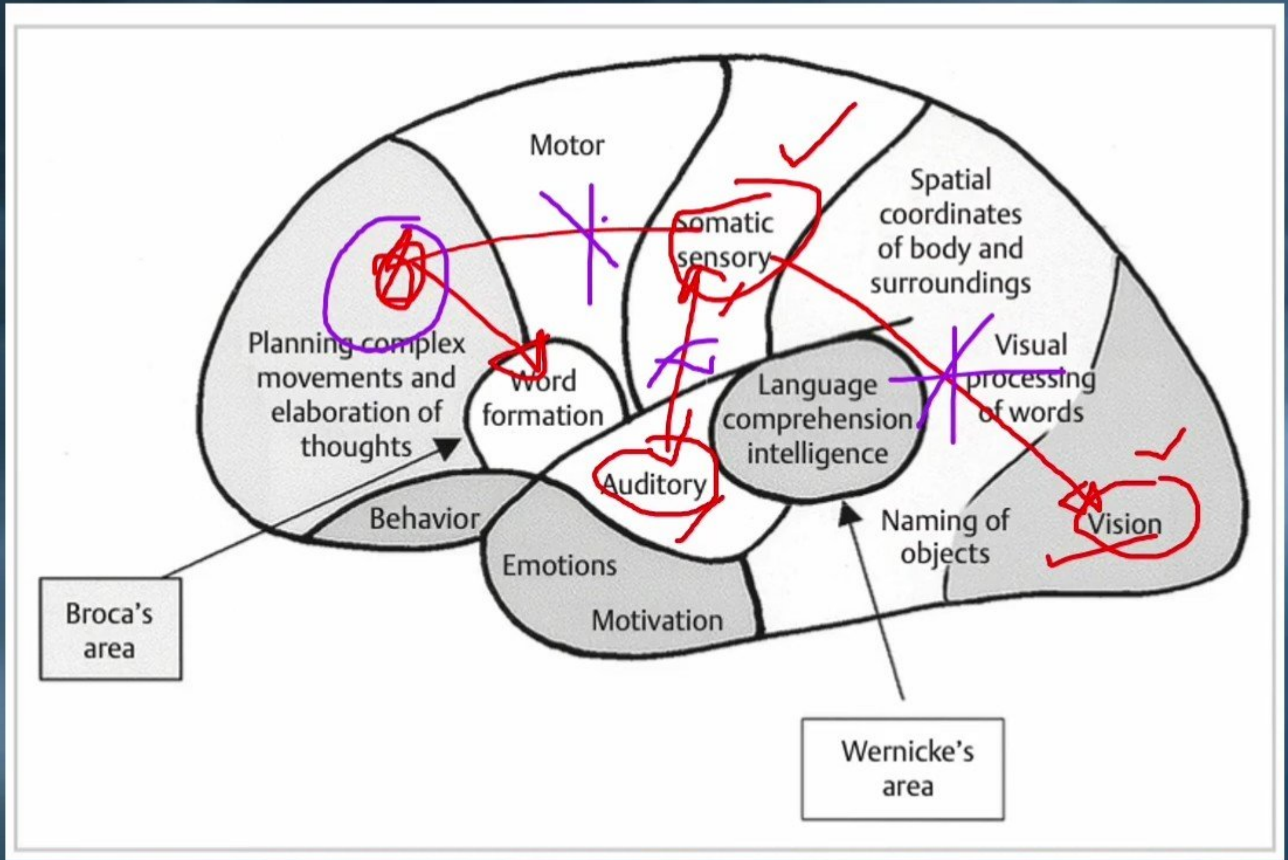
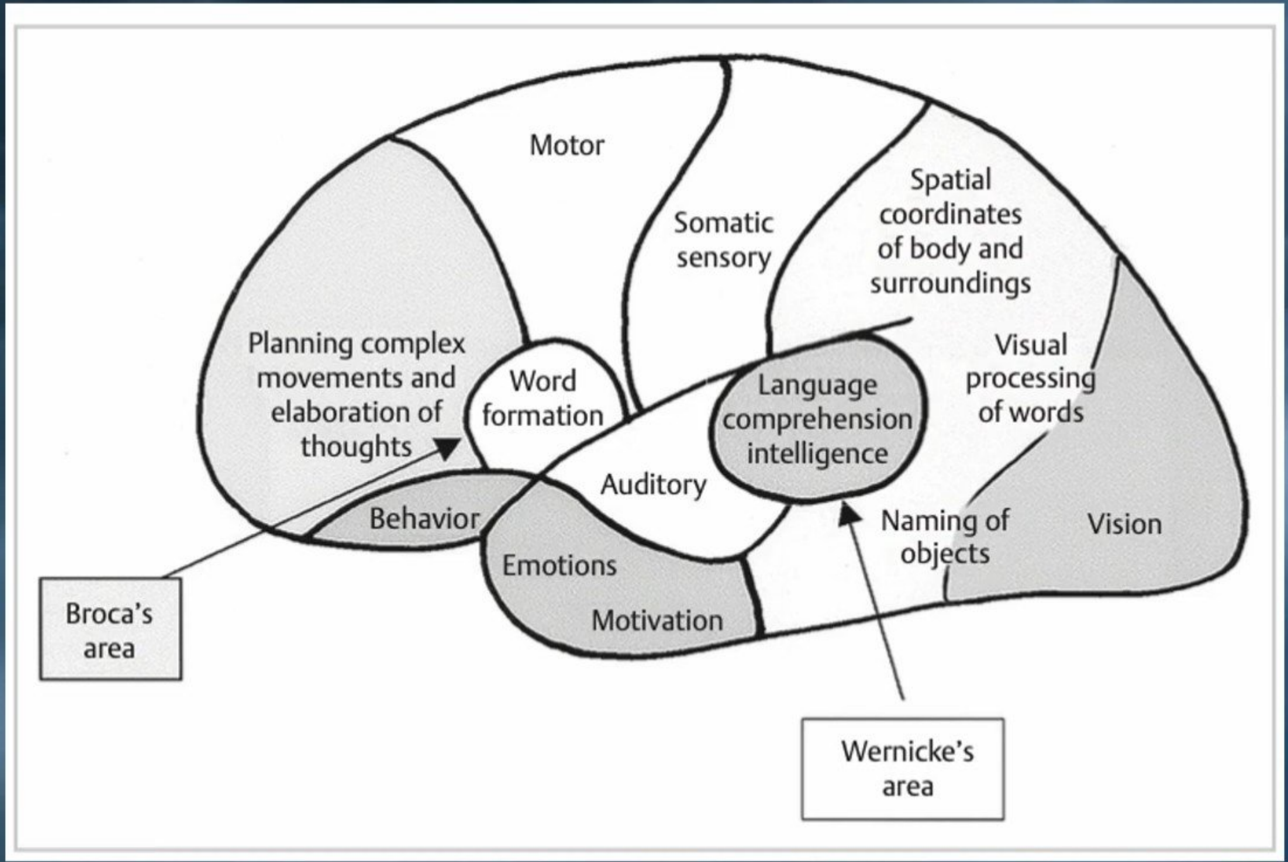


Association areas

1) Wernicke's area (Brodmann's areas 22, 39, 40):

- ✓ It is region of **language comprehension**.
- ✓ It is located behind the primary auditory cortex in the posterosuperior temporal lobe and inferior parietal lobe.
- ✓ It is the most important area for higher intellectual function because this is language based.
- ✓ It is found in **the left hemisphere** in virtually all right-handed individuals.
- ✓ **Stimulation** by tumors or trauma elicits complex thoughts, memory, visual scenes, and auditory hallucinations.



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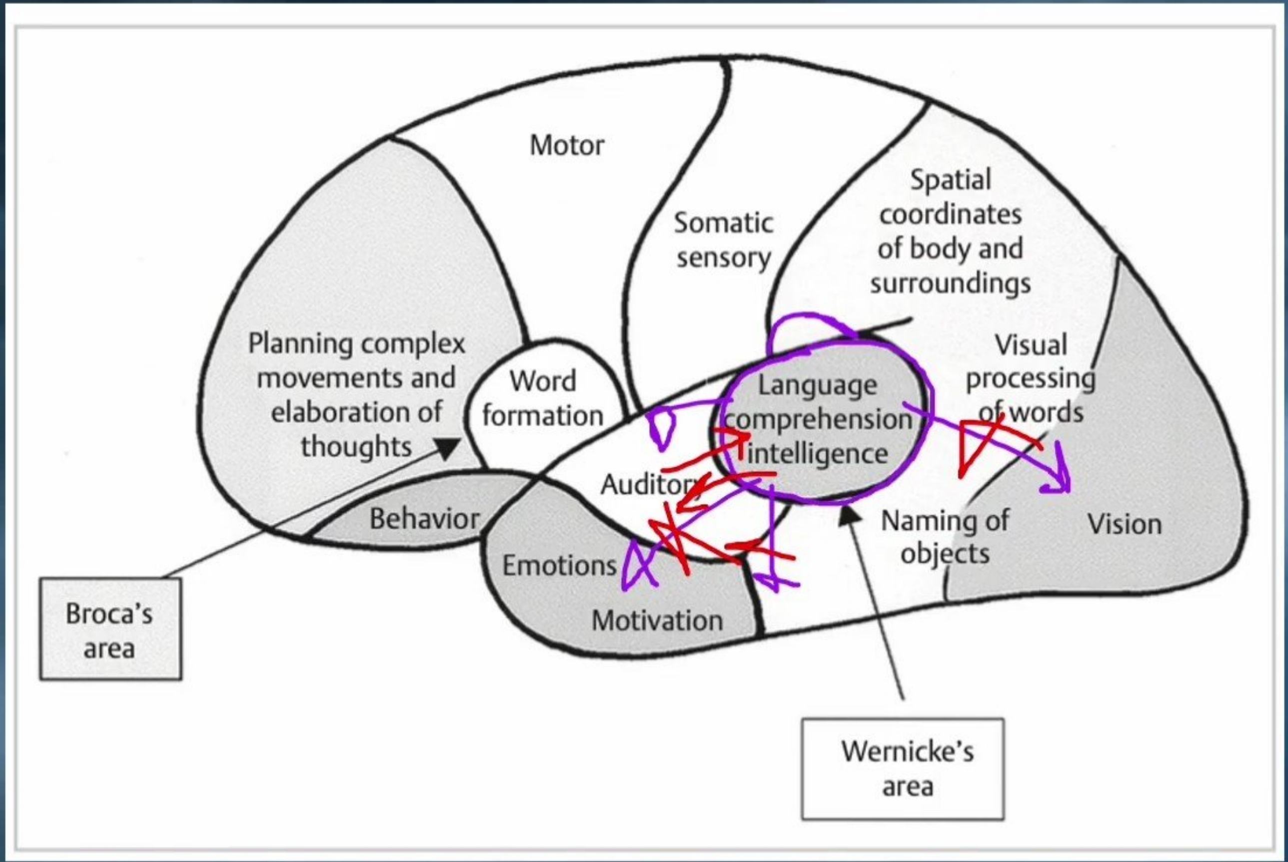


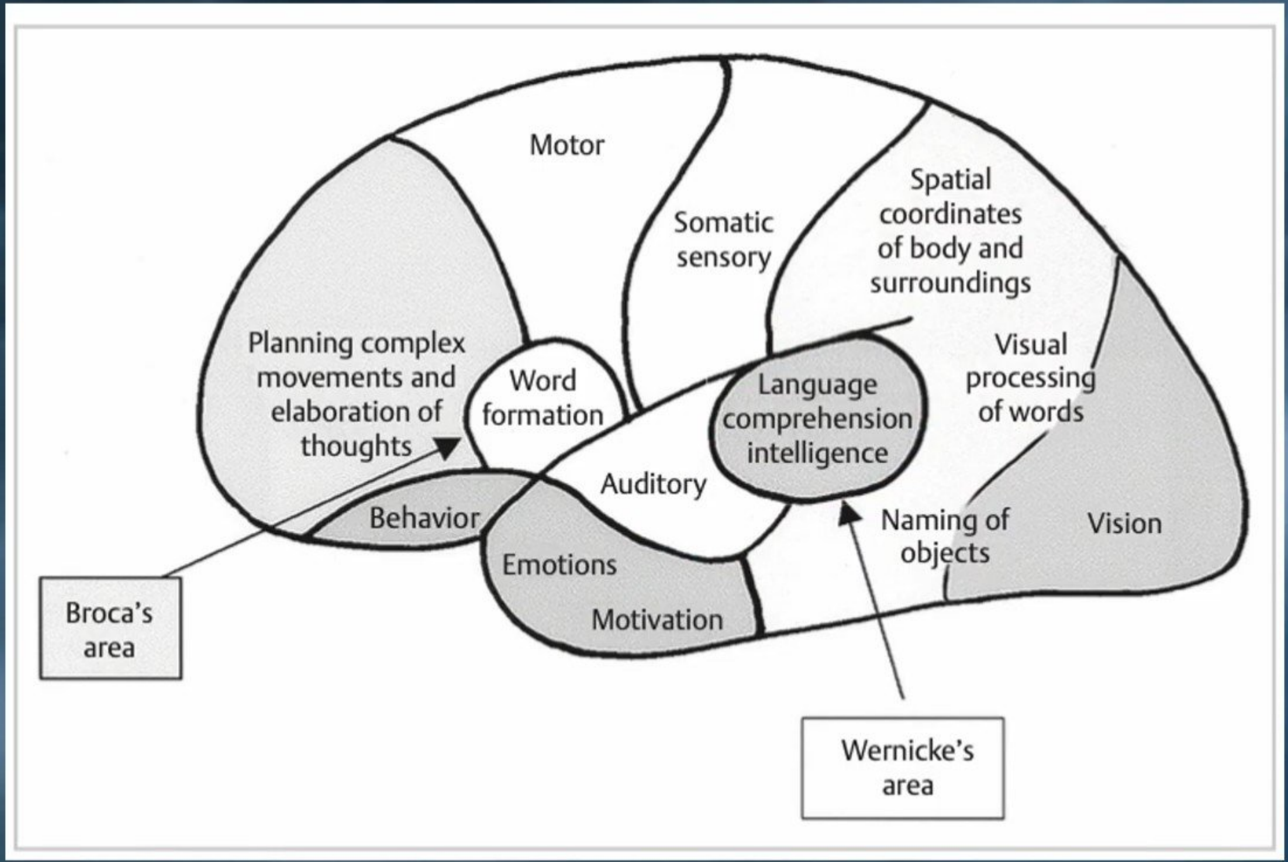
Association areas

2) Angular gyrus (Brodmann's area 39):

✓It lies in the posteroinferior parietal lobe and is responsible for the visual processing of words.

✓Damage to the angular gyrus produces dyslexia (inability to read) without a **deficit in understanding spoken language**.

✓Damage to the **auditory association area** results in word deafness. The ability to name a word is dictated by the lateral temporo- occipital junction.



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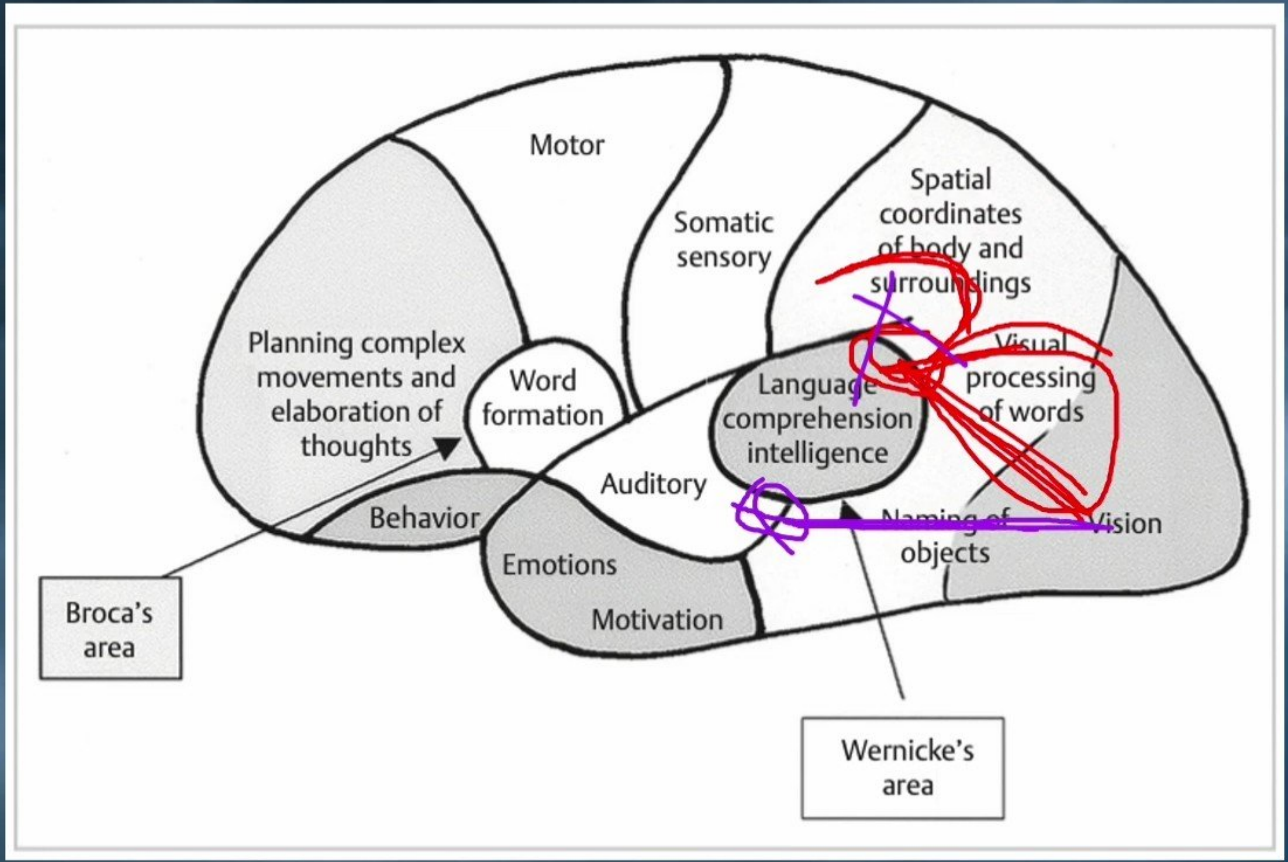
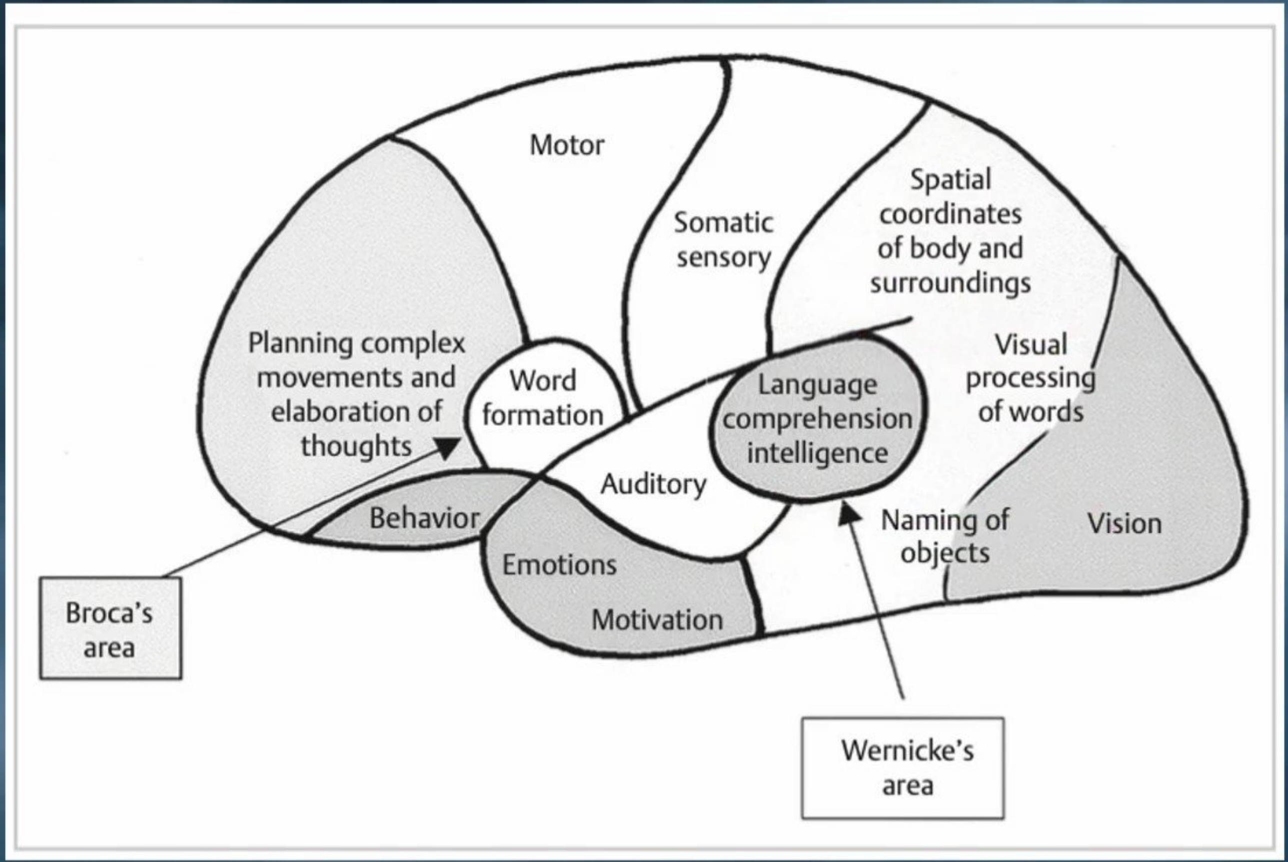


Association areas

3) Nondominant parieto-temporo-occipital cortex:

✓ It used for music, nonverbal visual expression, spatial relationships, body language, and voice intonations.



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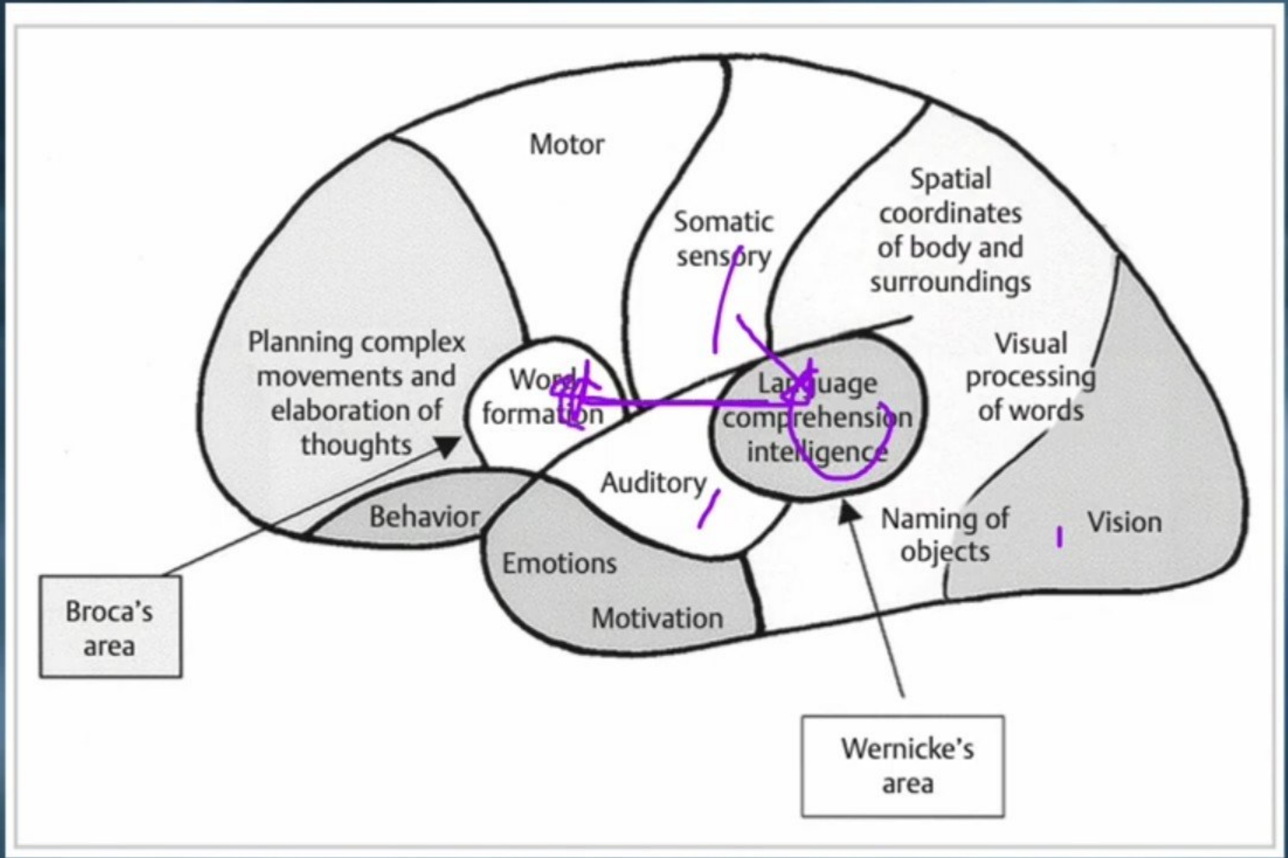
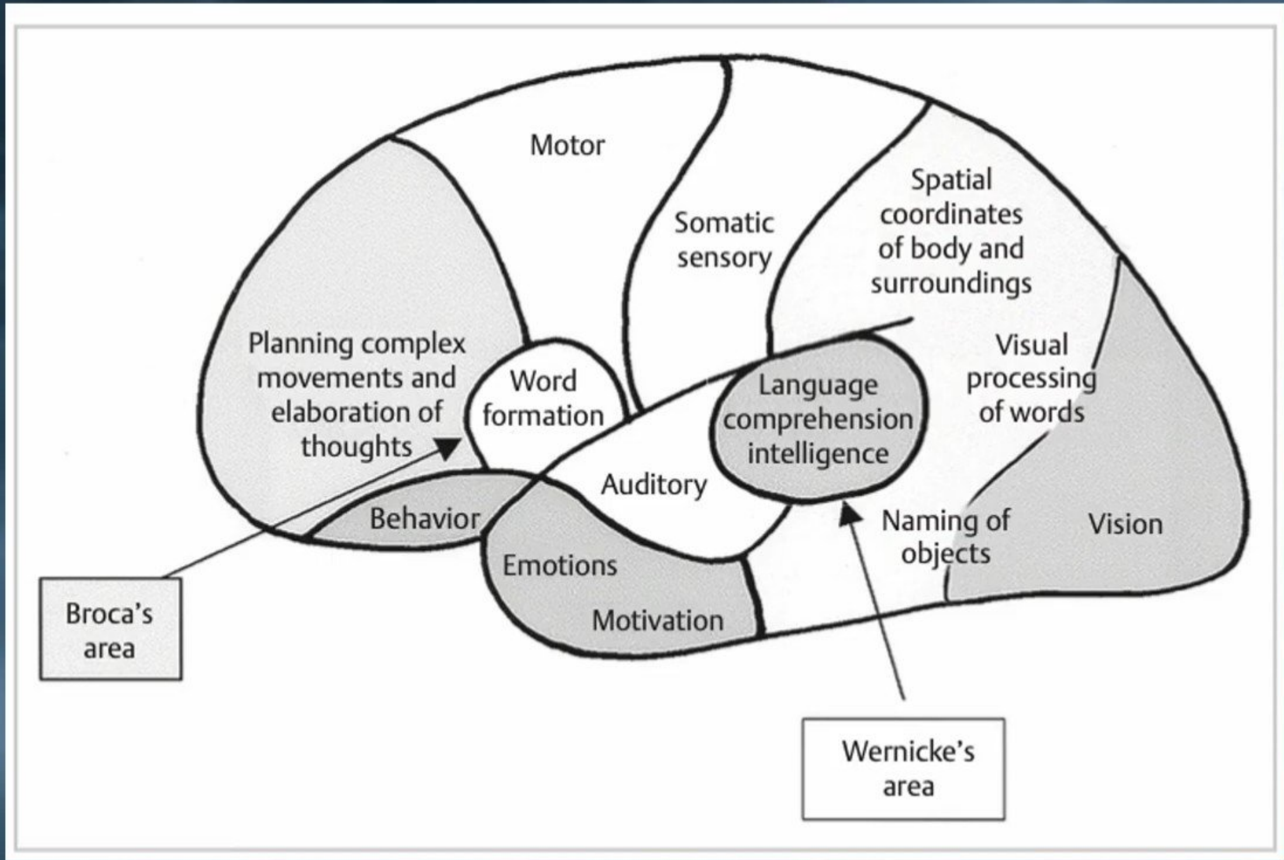


Association areas

B) Prefrontal association area:

- It coordinates with the **motor** cortex to plan complex patterns and sequences.
- It receives input from the parietal, temporal, and occipital association areas.
- Output is via the caudate loop for sequential and parallel movement complexes.



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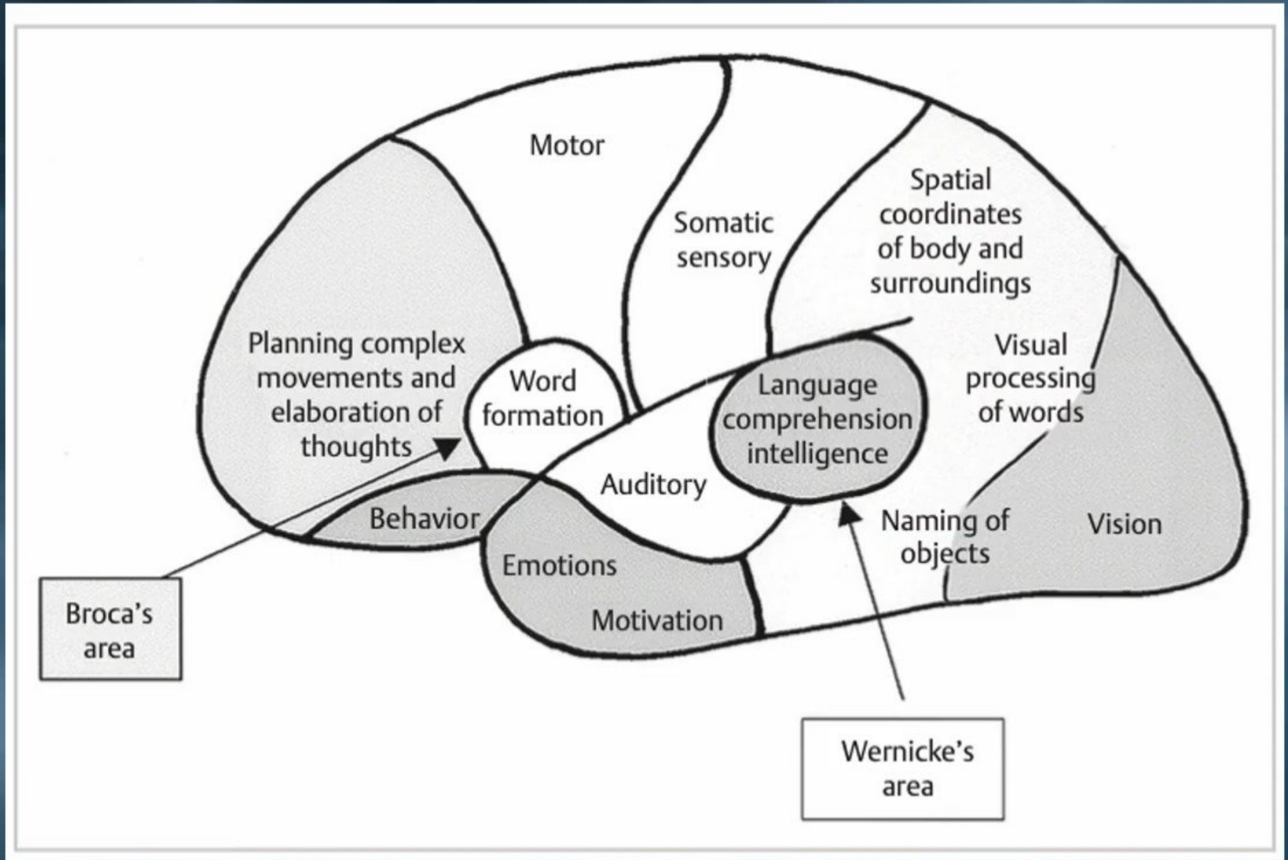
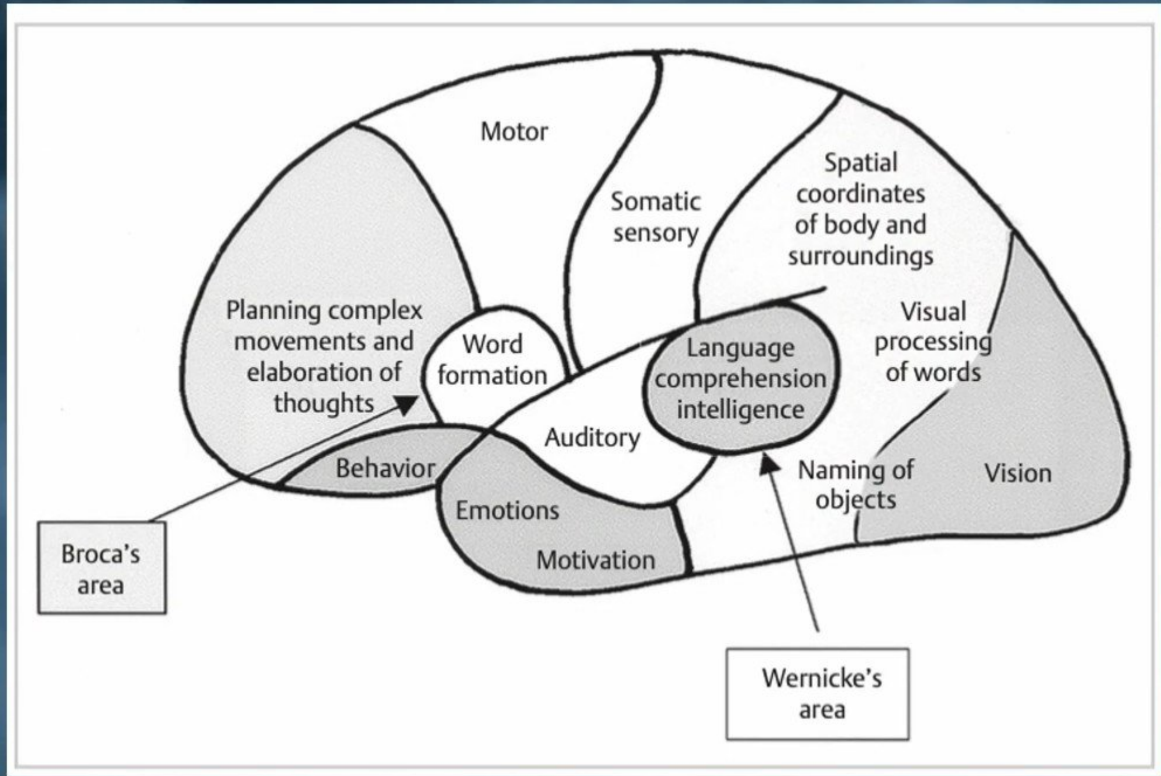


Association areas

1) Prefrontal cortex:

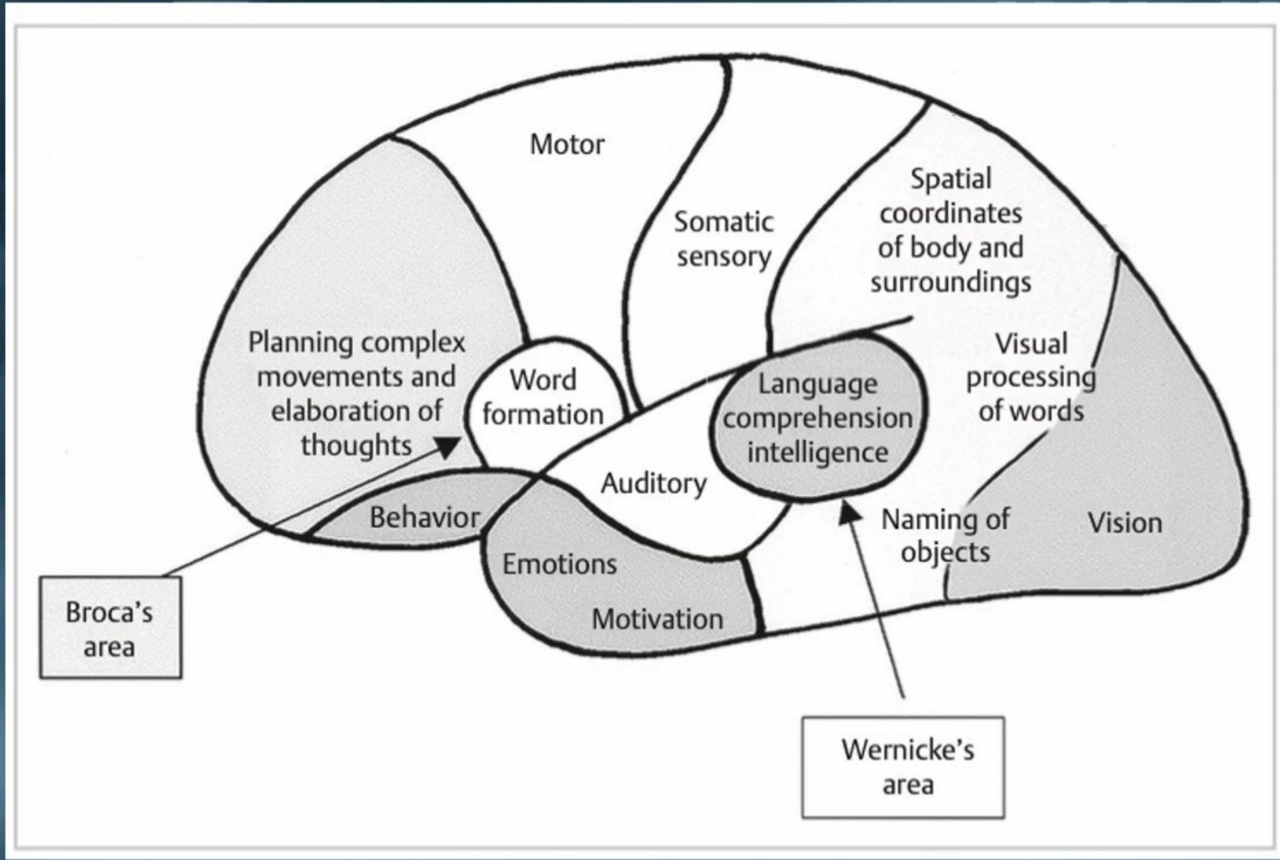
- It is used to prognosticate, plan, and delay action until all sensory input is considered, evaluates the consequences of actions, and solves complicated problems:
- After a **prefrontal lobotomy**, there is impaired complex problem solving, decreased ambition, decreased ability to perform sequential/parallel tasks, aggression (limbic system), decreased social responsiveness (especially with regard to sex and excretion), mood changes, decreased purpose, and decreased attention span.
- This is due to the fact that this cortex receives input from all areas and normally decides the appropriate motor response.



Association areas

2) Broca's area (Brodmann's area 44):

- ✓It controls word formation and **execution**.
- ✓It coordinates the simultaneous stimulation of respiratory, pharyngeal, and laryngeal muscles.
- ✓Part of it resides in the prefrontal cortex, and partly in the premotor cortex.



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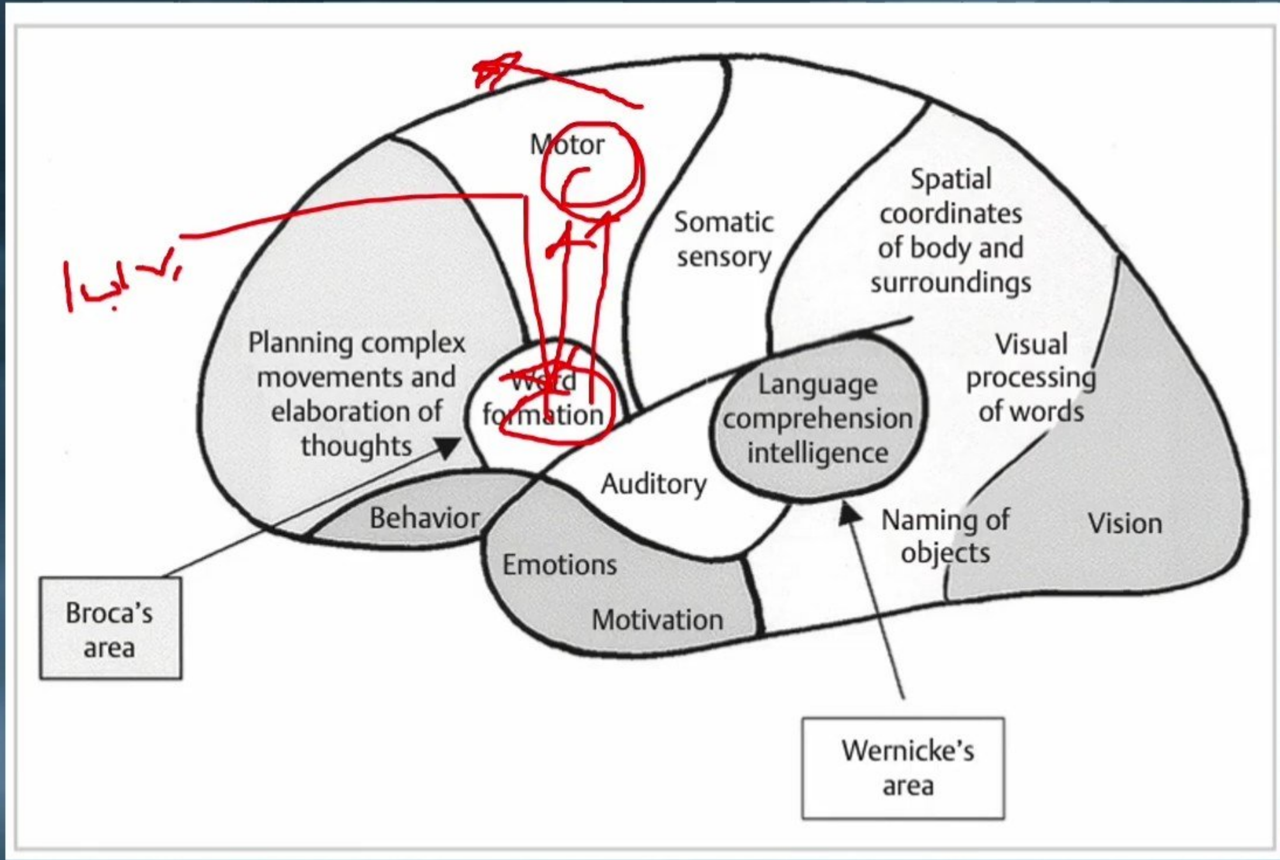


Association areas

C) Limbic association area:

✓It is located in **the antero-basal temporal lobe**, **basal frontal lobe**, and **cingulate gyrus.**

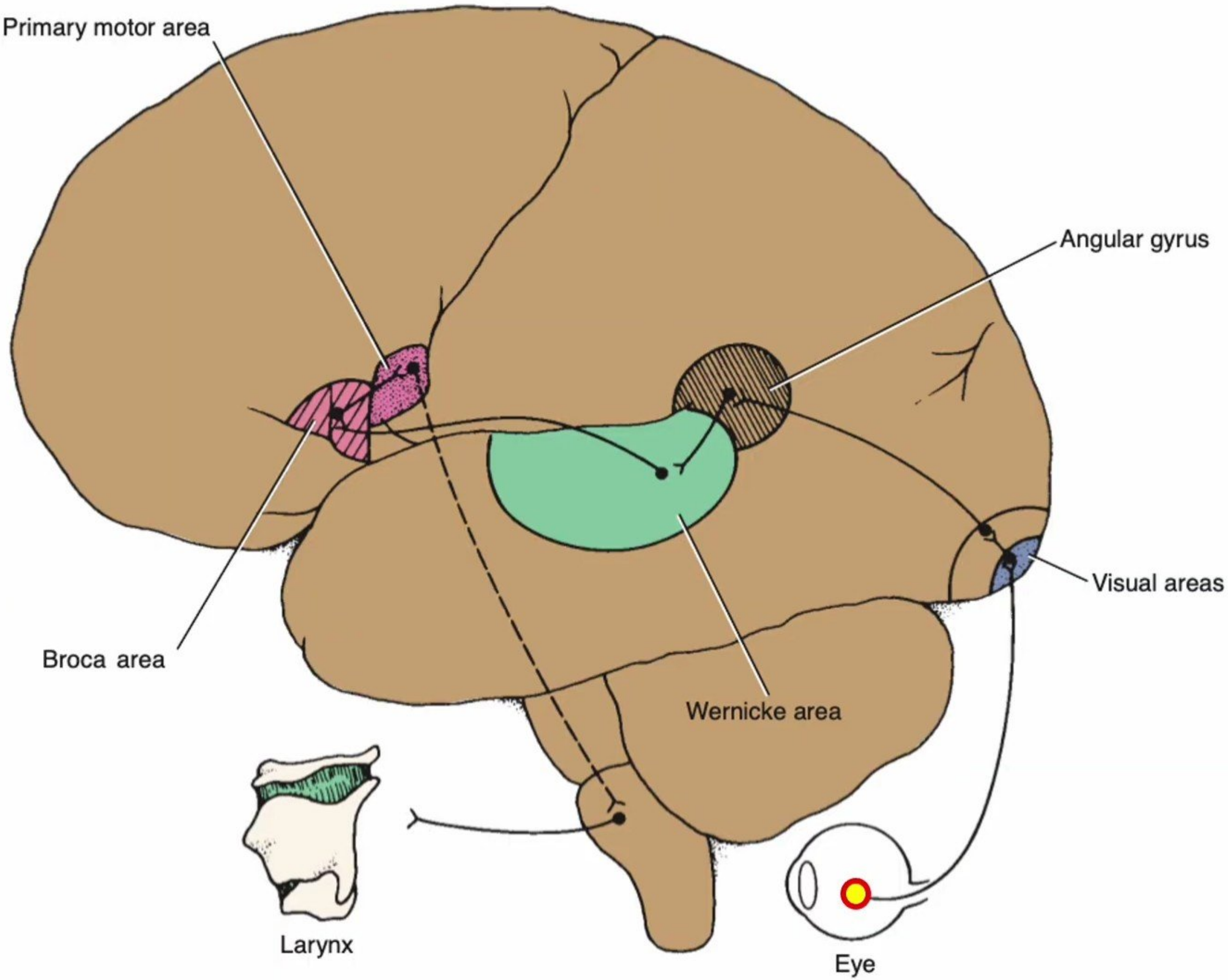
✓It is involved in behavior, motivation, and emotion.

D) Prosopagnosia:

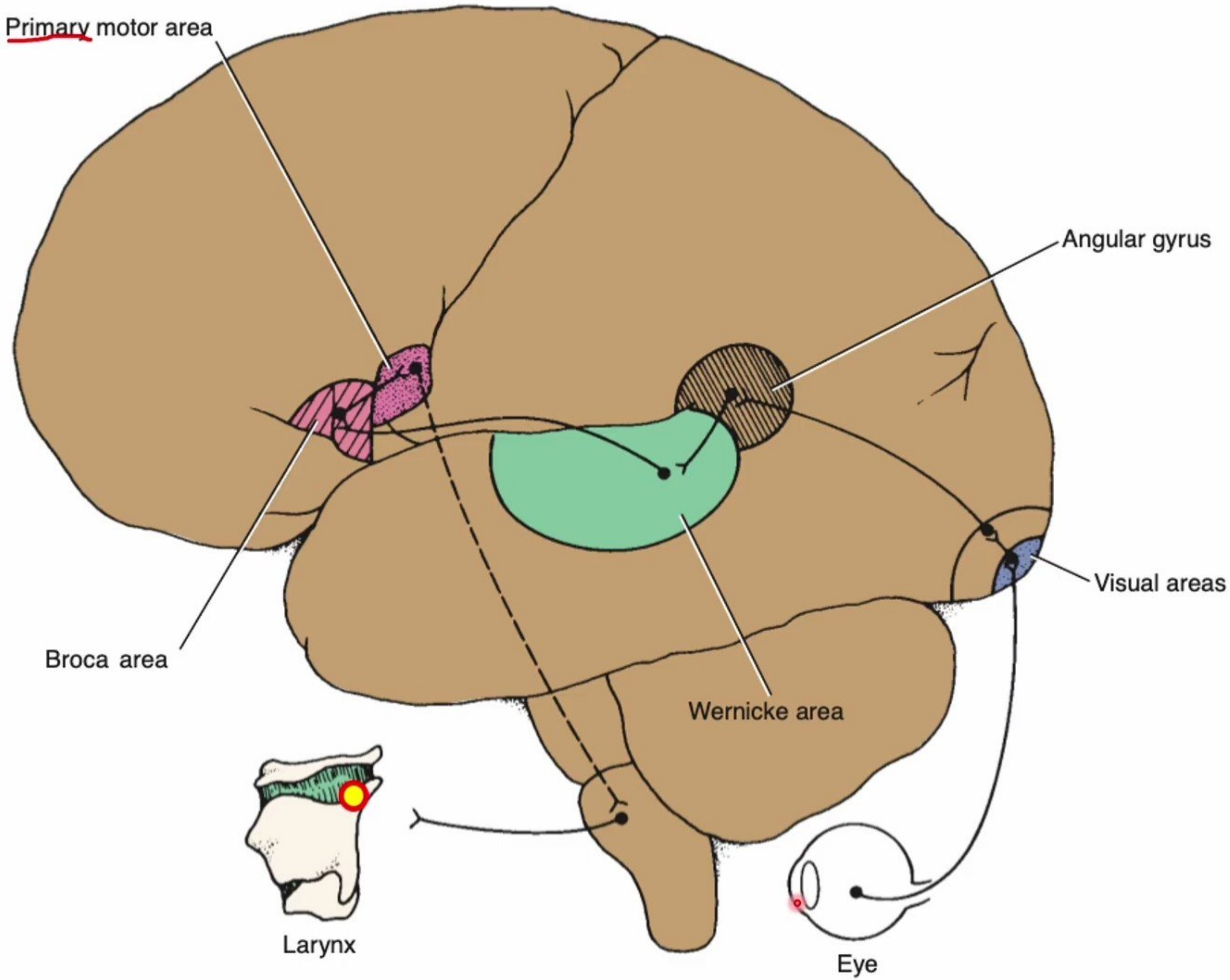
✓It is inability to recognize faces.

✓It is caused by **bilateral damage to the medial basal occipitotemporal cortex** between the limbic cortex in the temporal lobe and the visual cortex in the occipital lobe.

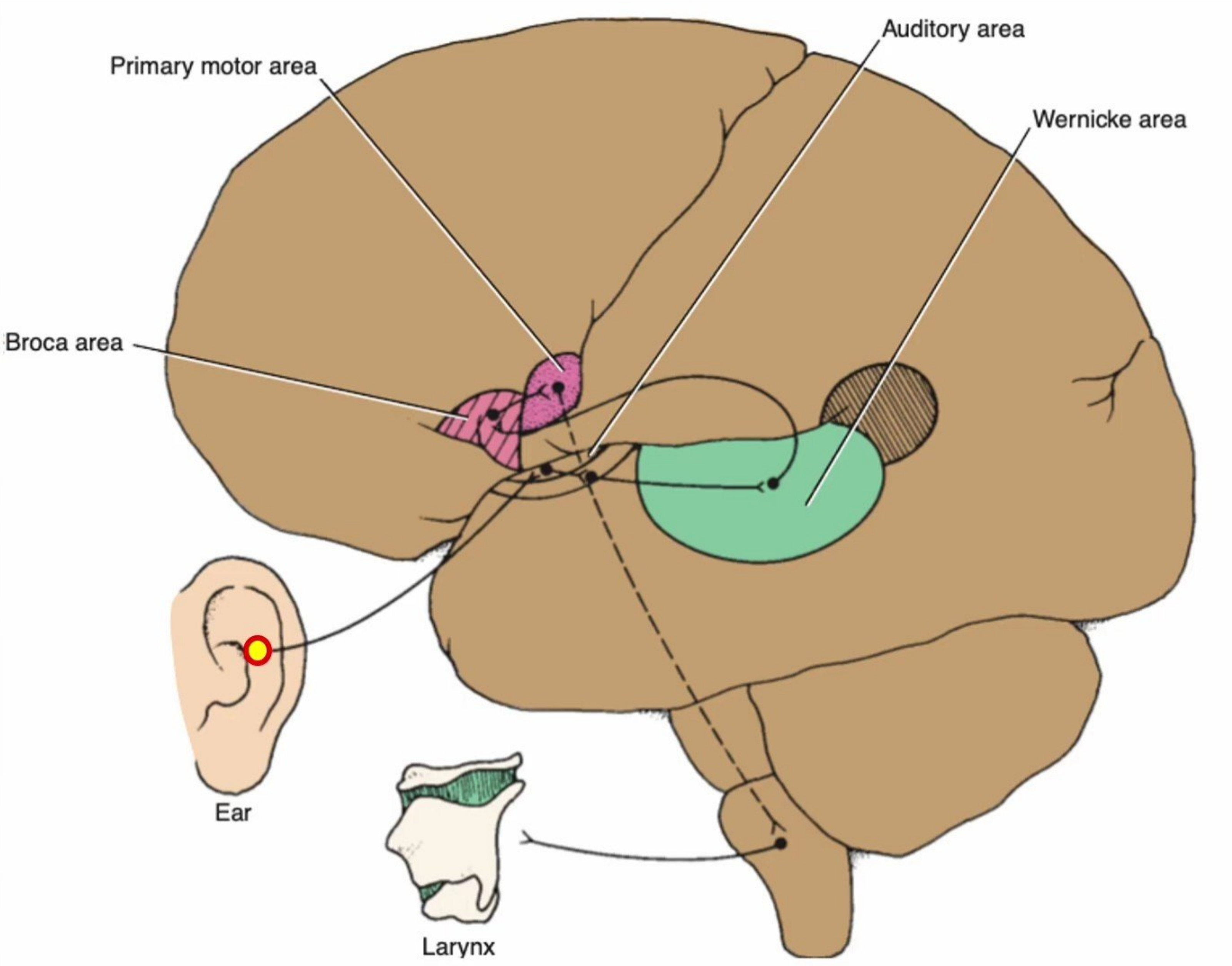
Probable nerve pathways involved in reading a sentence and repeating it out



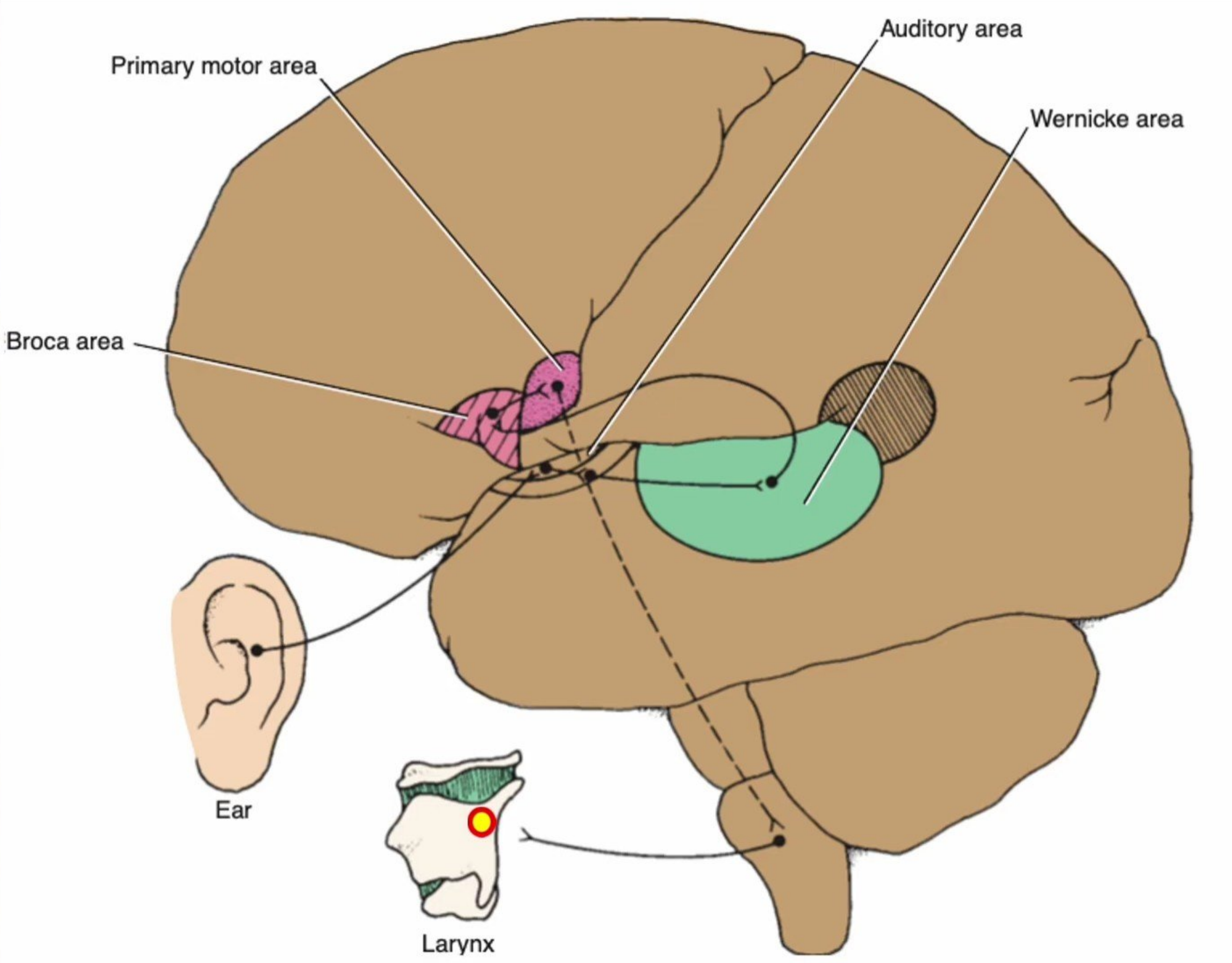
Probable nerve pathways involved in reading a sentence and repeating it out



Probable nerve pathways involved with hearing a question and answering it



Probable nerve pathways involved with hearing a question and answering it




Thanks